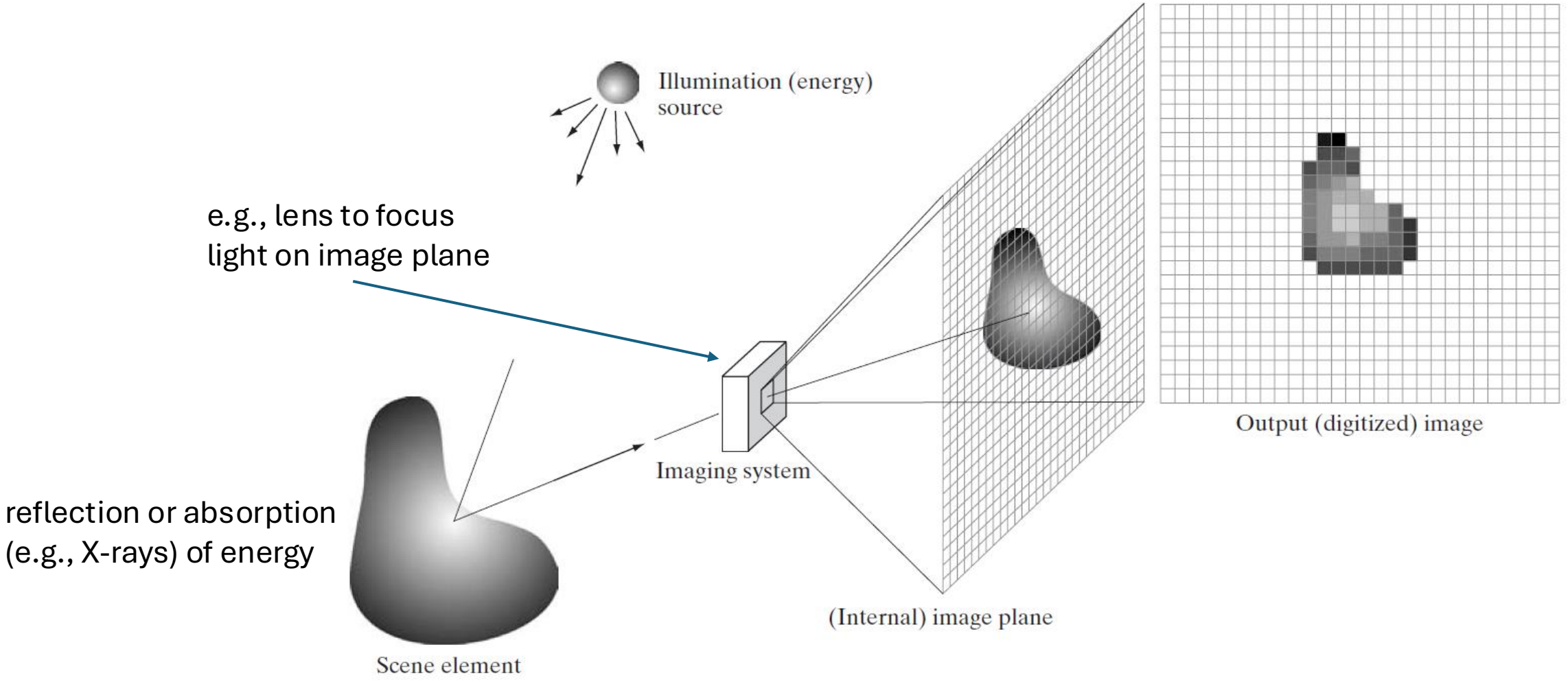
Digital Image Processing

Computer Vision

Digital Image Acquisition Process



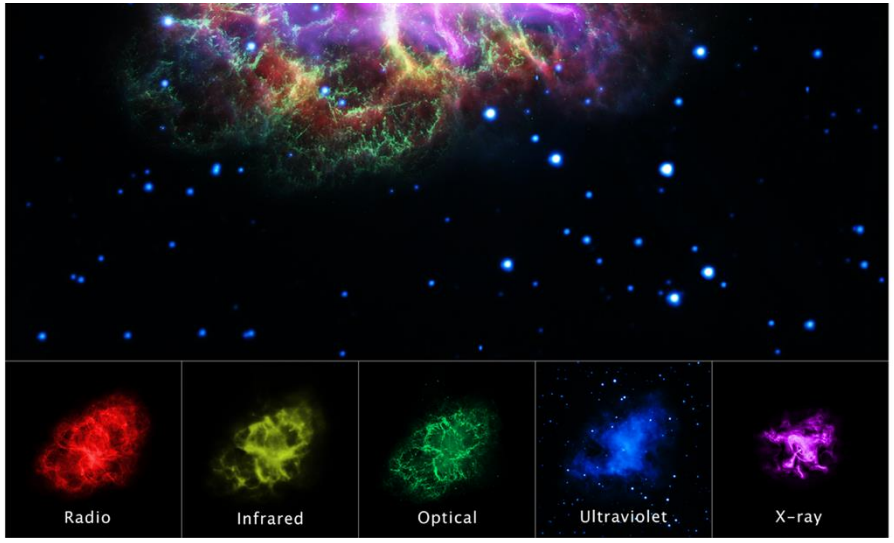
Energy Sources

Electromagnetic Spectrum

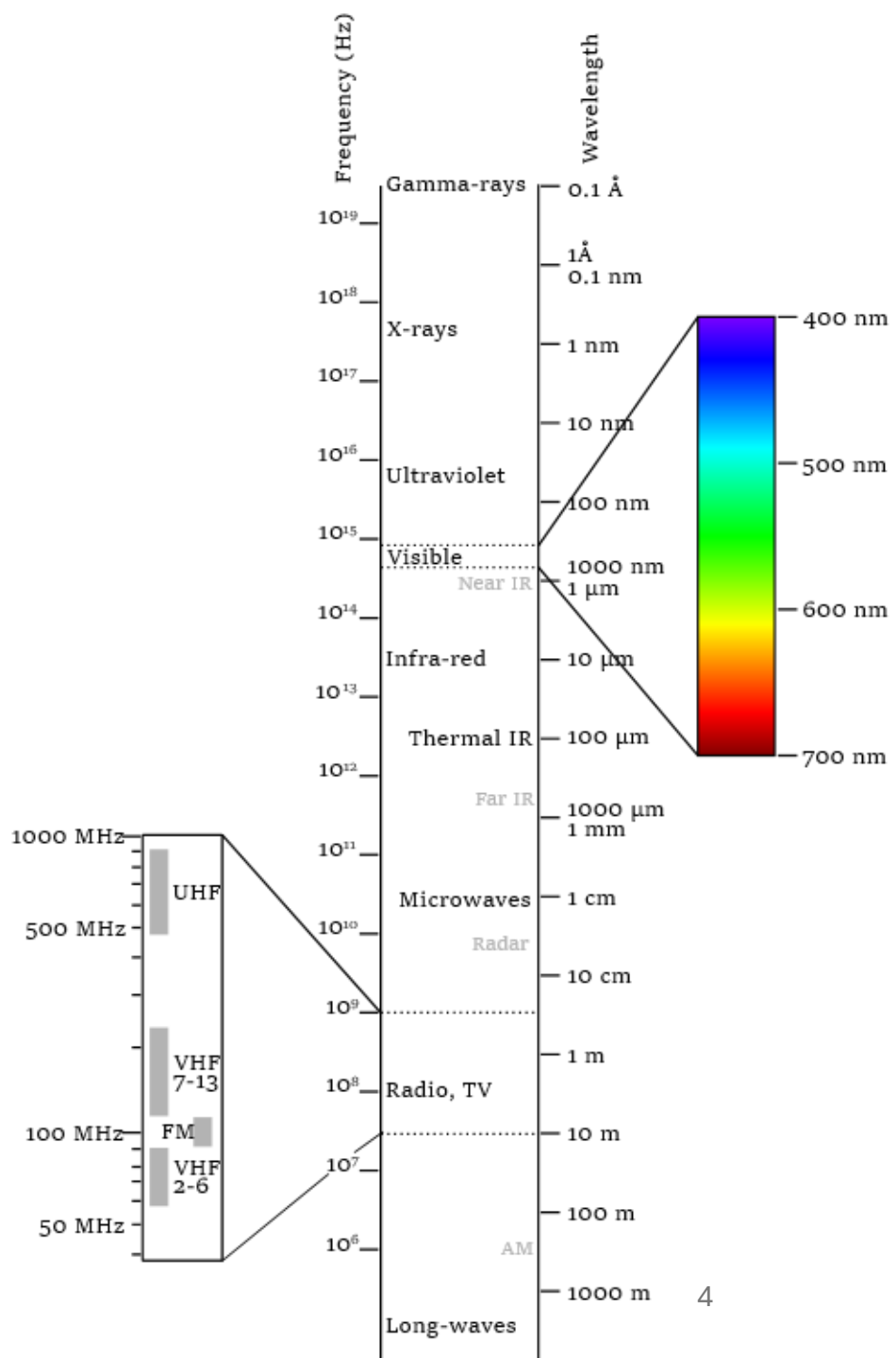
examples:

- visible band: digital camera
- microwaves: radar
- X-rays: computed tomography (CT)
- radio band: magnetic resonance imaging (MRI)
- gamma rays: positron emission tomography (PET)

Crab Nebula in multiple wavelengths



source



Lidar vs Cameras for Self-Driving Cars

Lidar:

- similar to radar, but using lasers
- very accurate



camera:

- human-like perception
- cost-effective

Examples for other Energy Sources

ultrasound



electron beams
(electron microscopy)

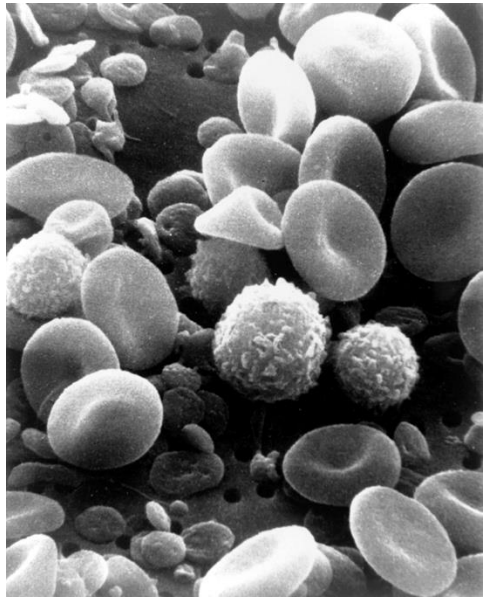
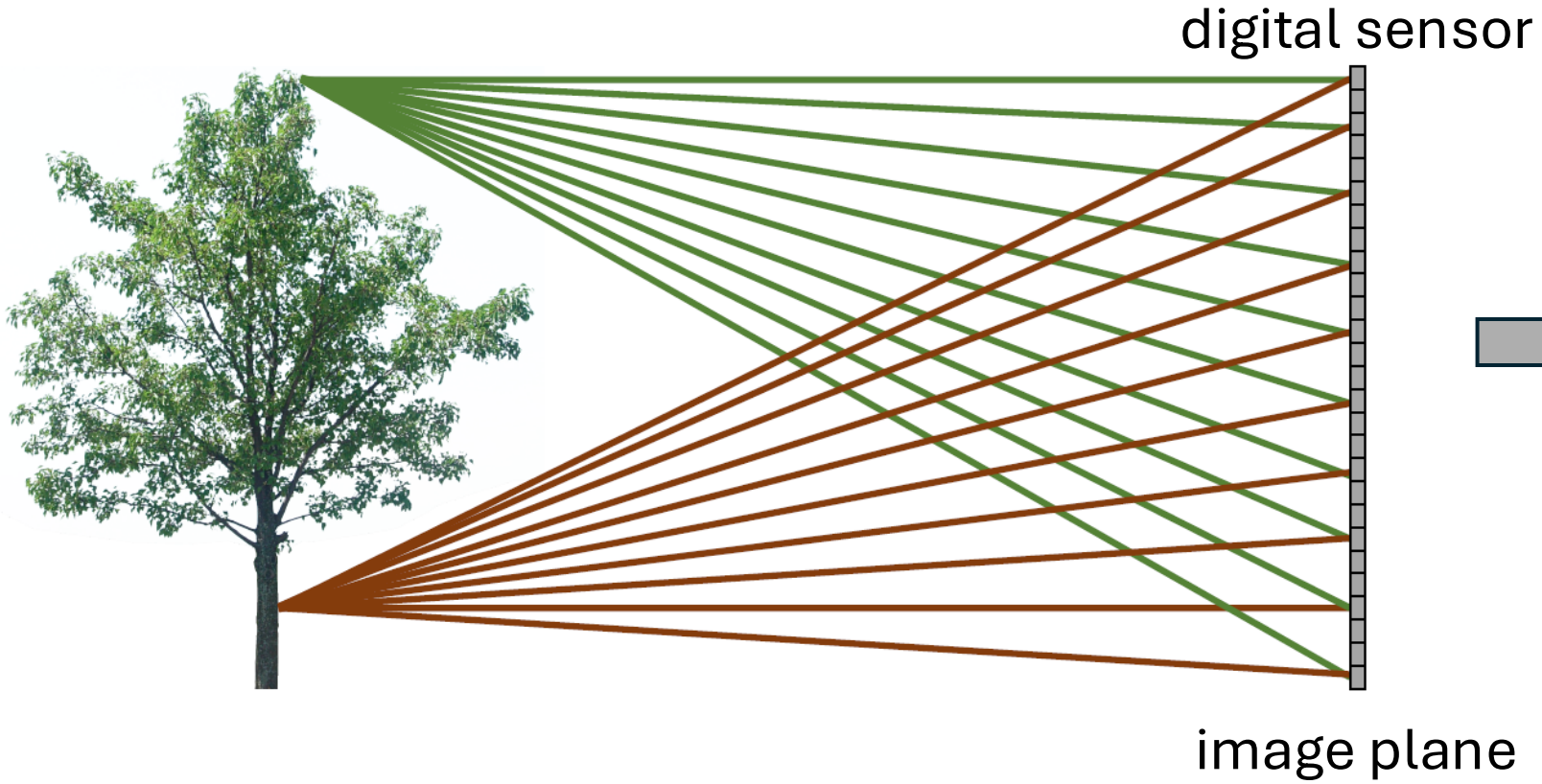


Image Formation

Bare-Sensor Imaging

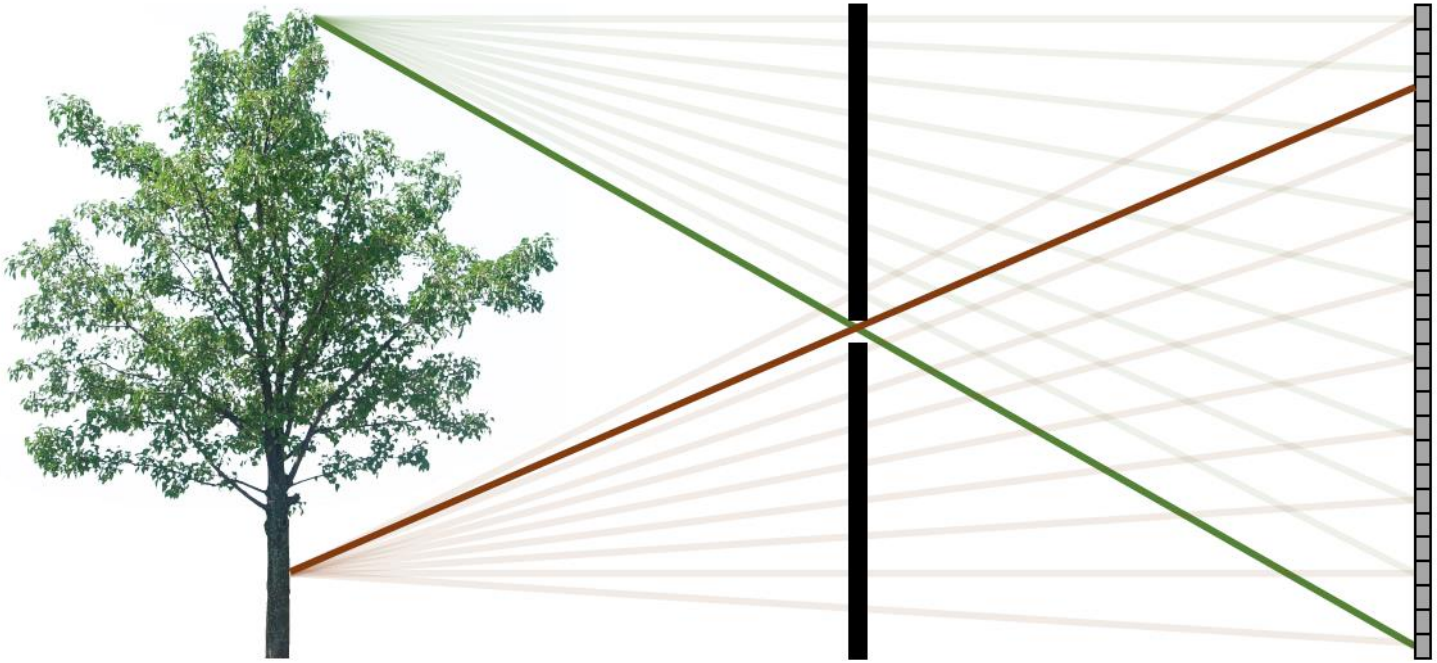


all scene points contribute to all sensor pixels:

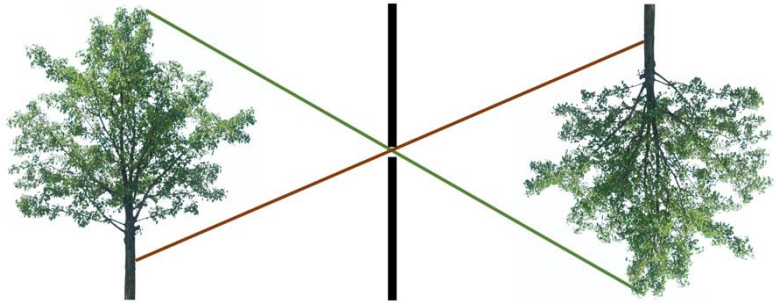


Pinhole Camera

camera: barrier with
pinhole (aperture)

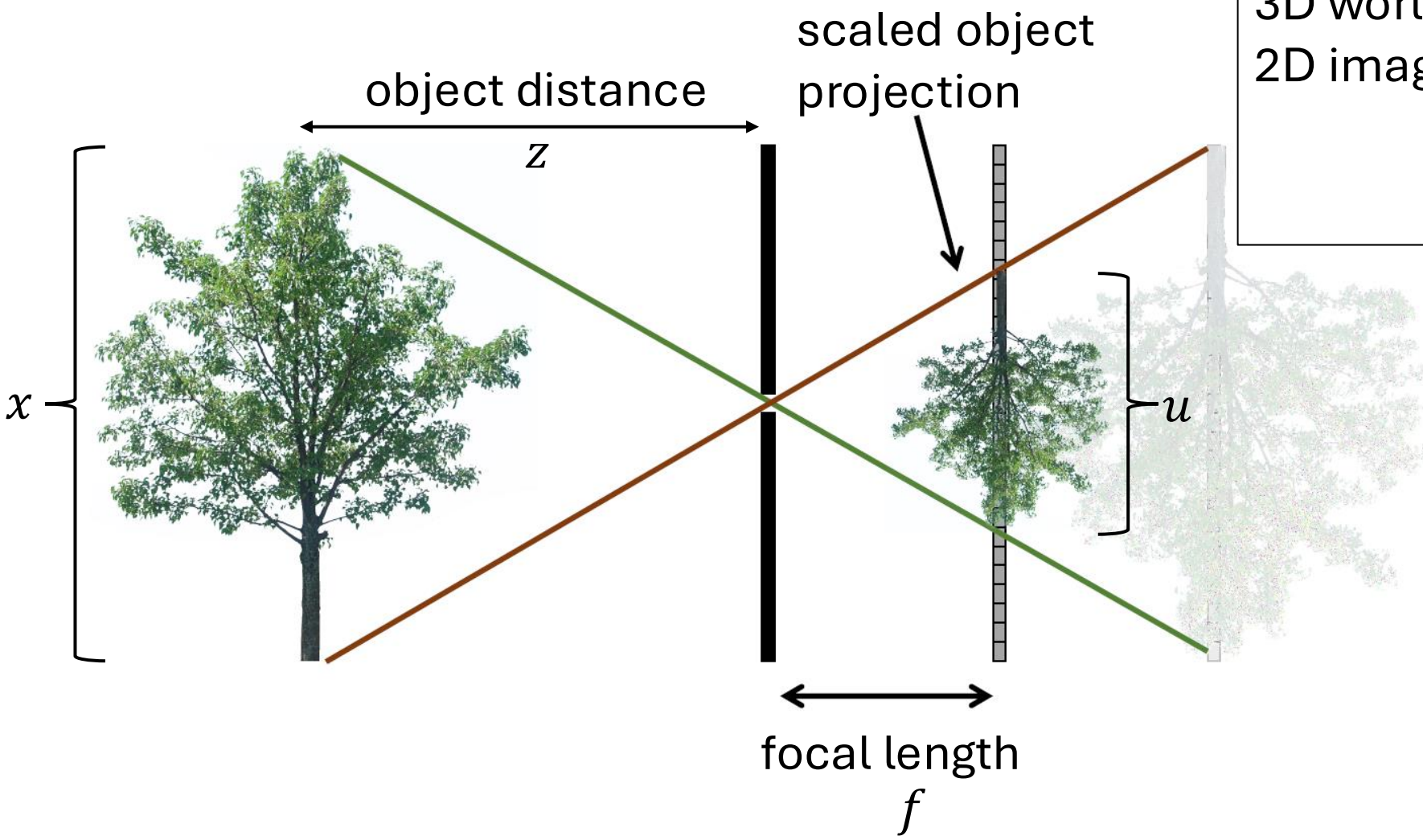


camera obscura



each scene point contributes
to one pixel only

Pinhole Camera Model



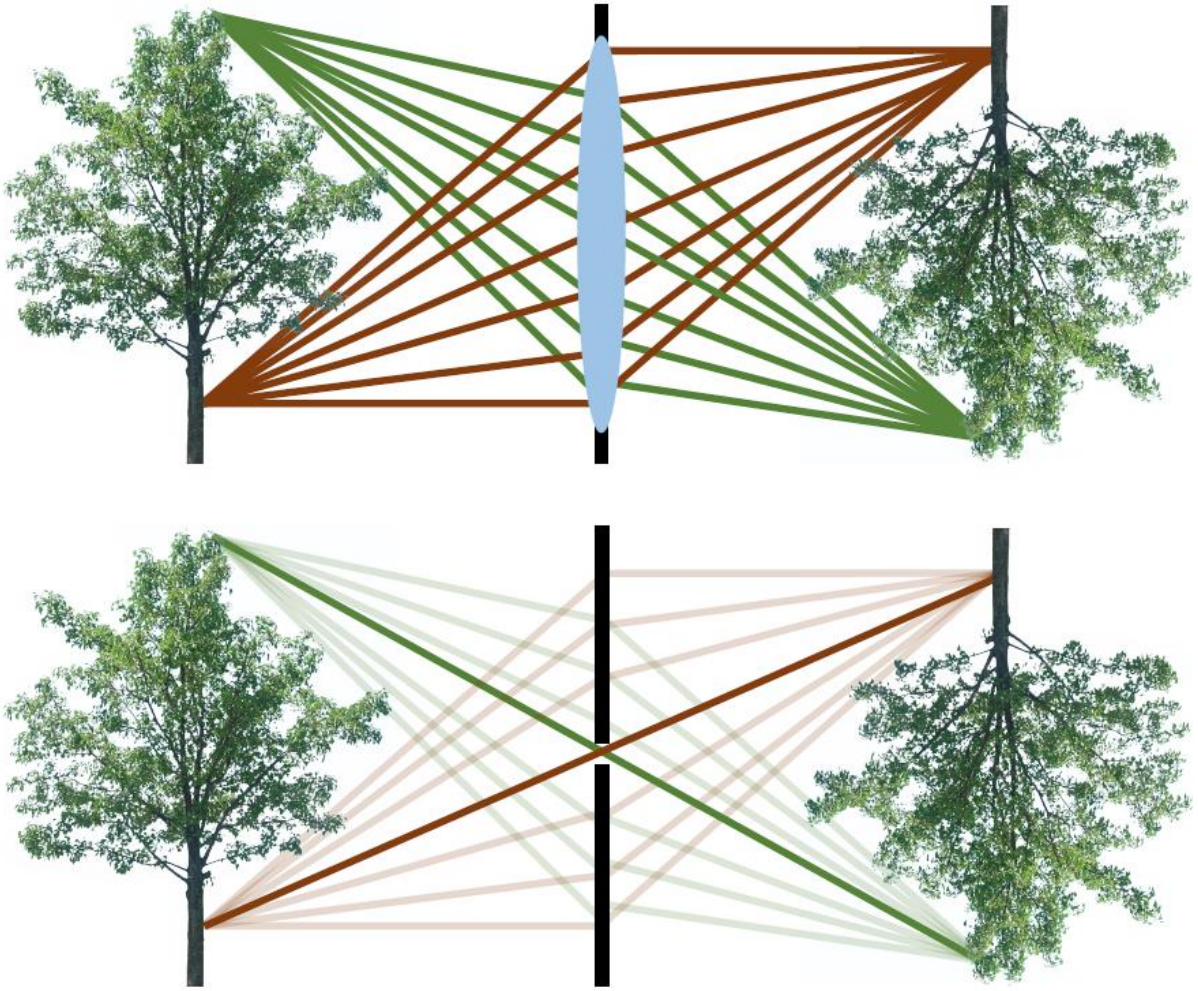
3D world coordinates (x, y, z) to 2D image coordinates (u, v):

$$\begin{pmatrix} u \\ v \end{pmatrix} = -\frac{f}{z}\begin{pmatrix} x \\ y \end{pmatrix}$$

losing depth …



Lens Camera

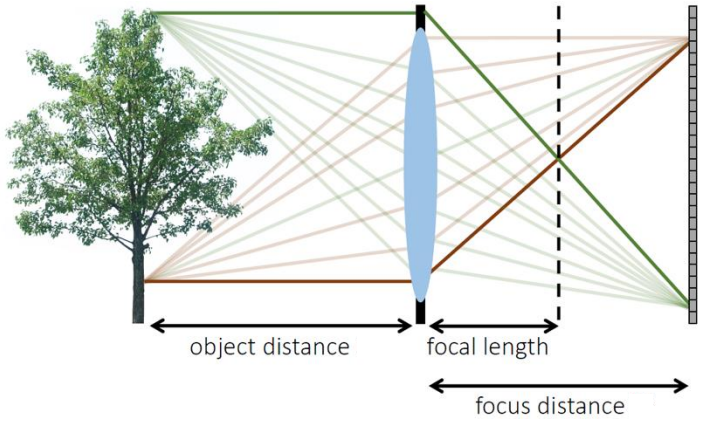


ideal pinhole is infinitesimally small: larger diameter/aperture makes image blurrier (but also allows more light to pass through: higher intensity of image pixels)

→ use optical lens

lens camera can be described with pinhole camera model, if:

- only central rays are considered
- lens camera is in focus (focus distance of lens camera equal to focal length of pinhole camera)



Digital Image Sensors

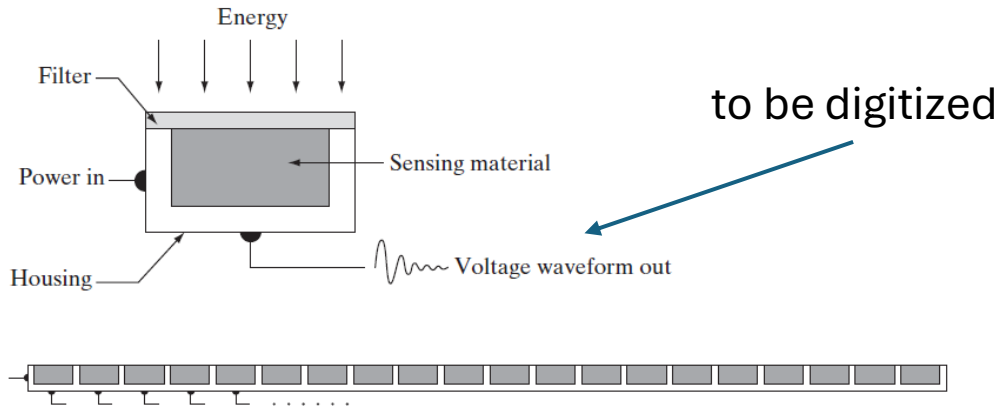
Photons to Electrons

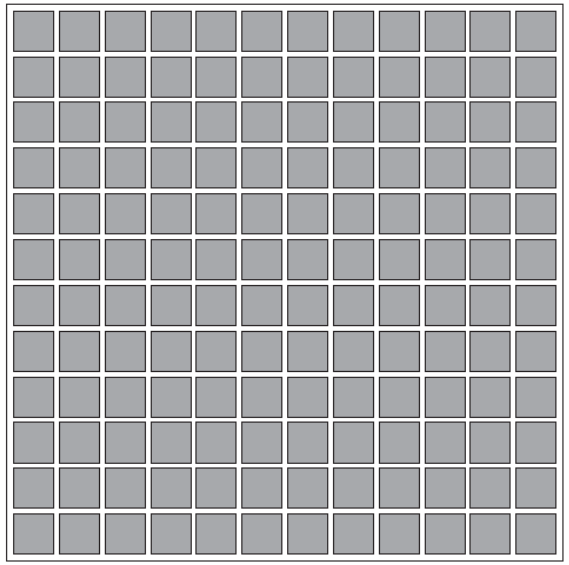
largely replaced chemical and analog imaging

single sensor: photodiode

sensor strips: motion perpendicular to strip for imaging in other direction, used in CT

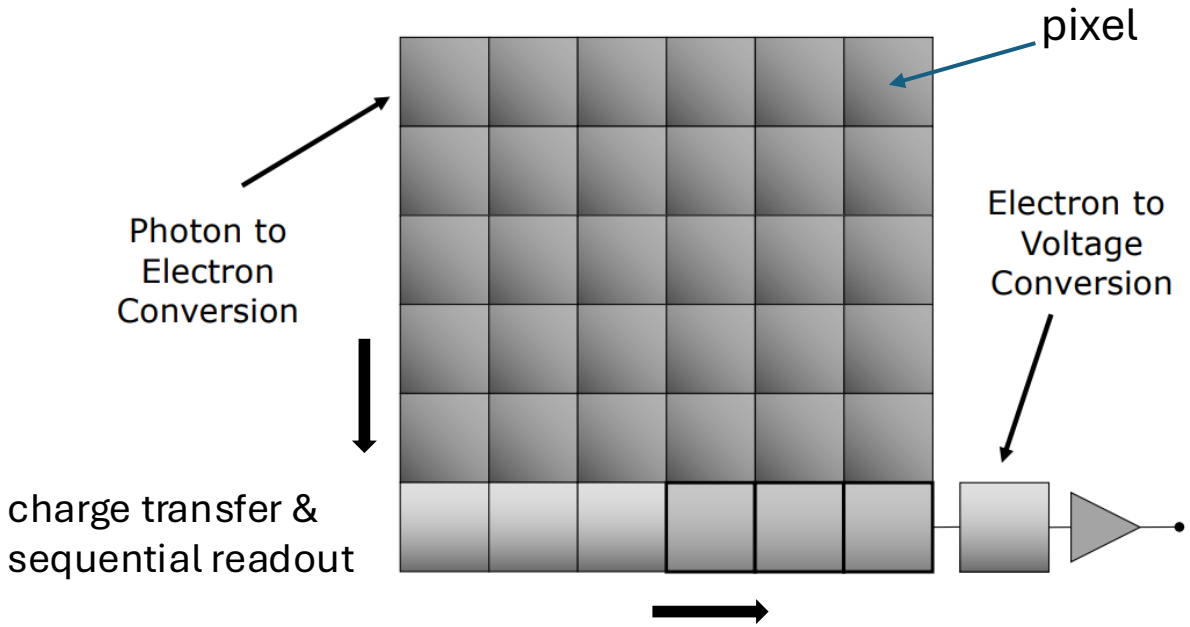
sensor arrays: used in digital cameras (in focal plane, outputs proportional to integral of light received at each sensor)





Most Popular Sensors: CCD and CMOS

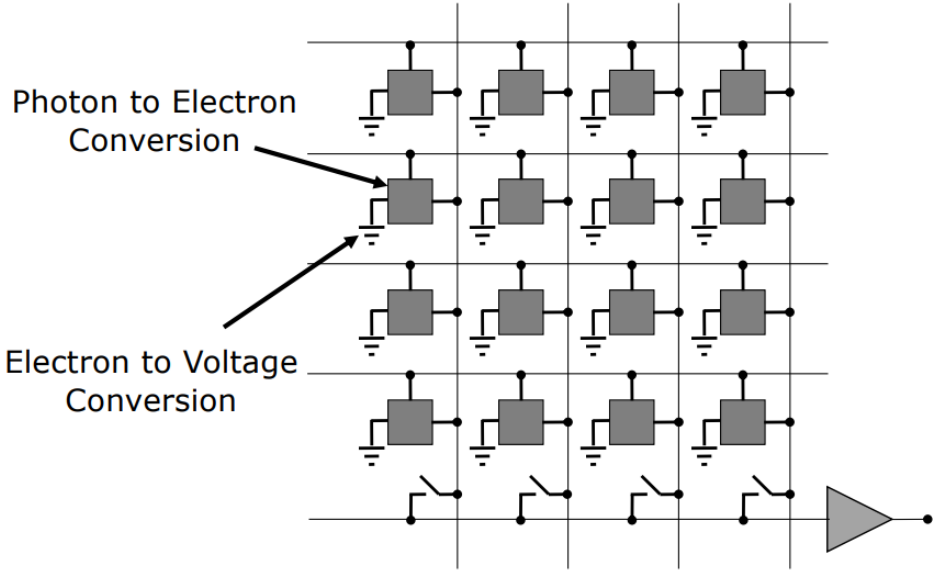
CCD: Charge Coupled Device



larger light-sensitive areas, lower noise level
→ used in high-end video cameras

active pixel sensor

CMOS: Complementary Metal-Oxide Semiconductor



lower power consumption, typically smaller
→ used in most consumer product cameras

no individual silicon cells, but microlenses on top of each pixel to focus light on sensitive areas

Global Shutter CMOS



in fact, no individual channels per CMOS-pixel: for reasons of space, pixels share lines and are read out row by row (rolling shutter)

→ potential skew (top part of the image captured slightly earlier than bottom part)

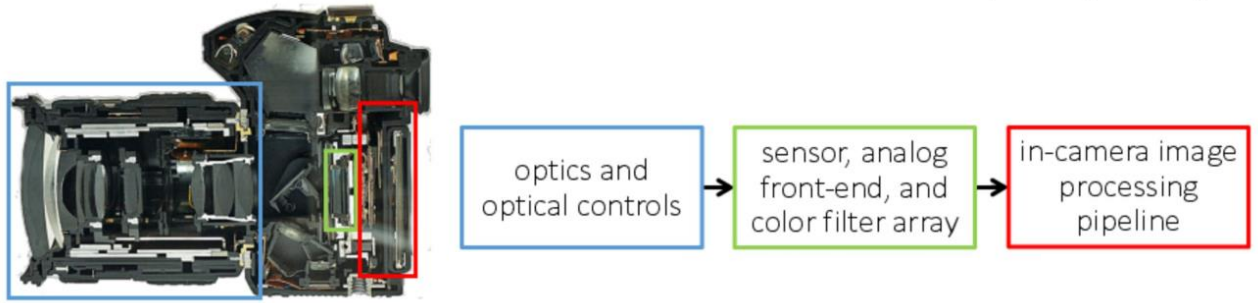
(more expensive) solution: global shutter

in-pixel storage enables simultaneous capture: all pixels expose together and instantly freeze data in tiny analog buffers, allowing row-by-row readout without motion distortion

global shutter CMOS replaces legacy CCDs: matches distortion-free motion tracking while drastically outperforming CCD in speed, power efficiency, and video resolution

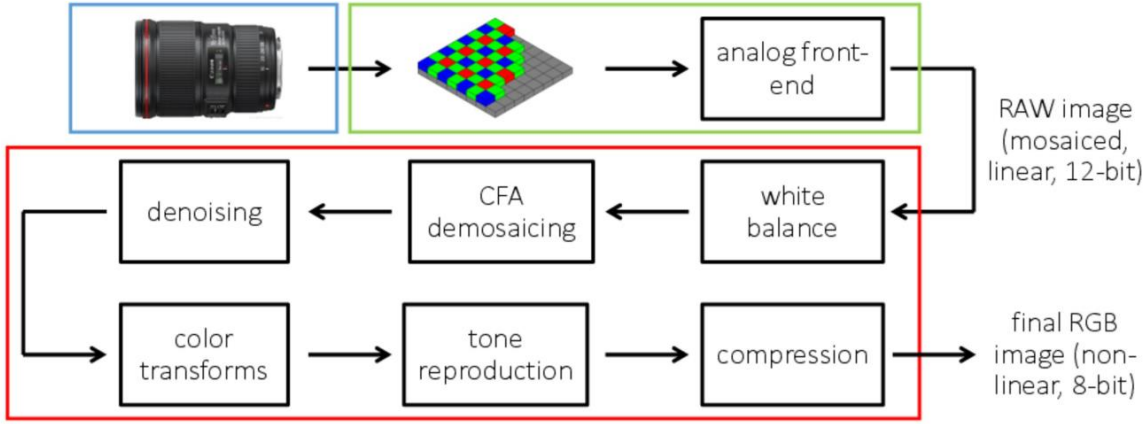
The modern photography pipeline





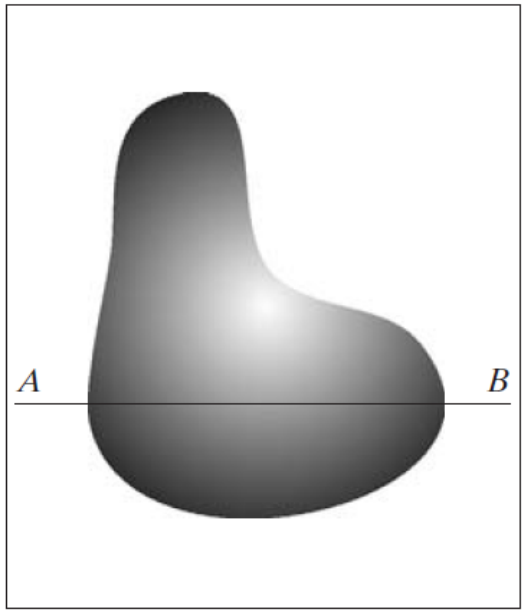
1. camera shutter opens
2. photodiodes: photons to electrons, store in potential well
3. camera shutter closes

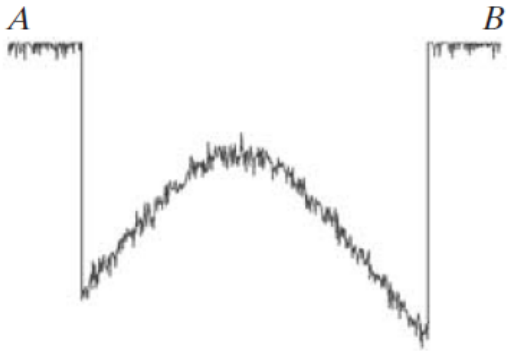
4. readout of potential wells, conversion to analog signals
5. conversion to digital signals (intensity levels)

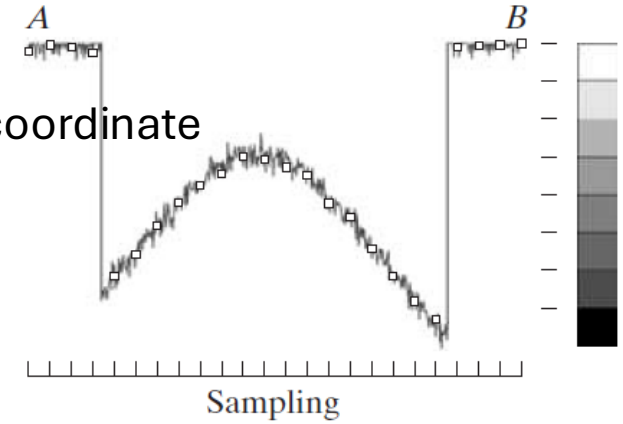


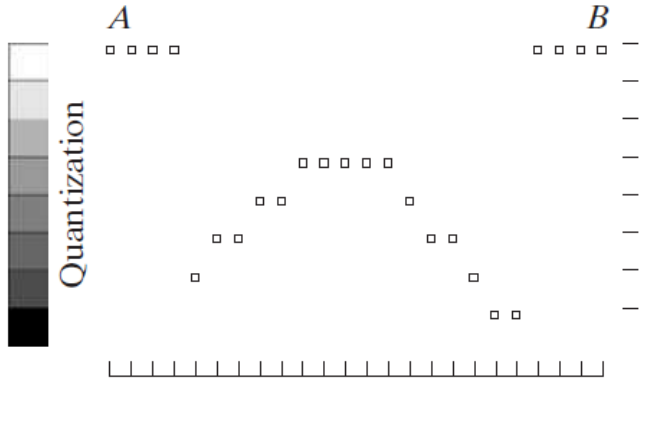
Pixels & Intensity

Digitizing: Sampling & Quantization

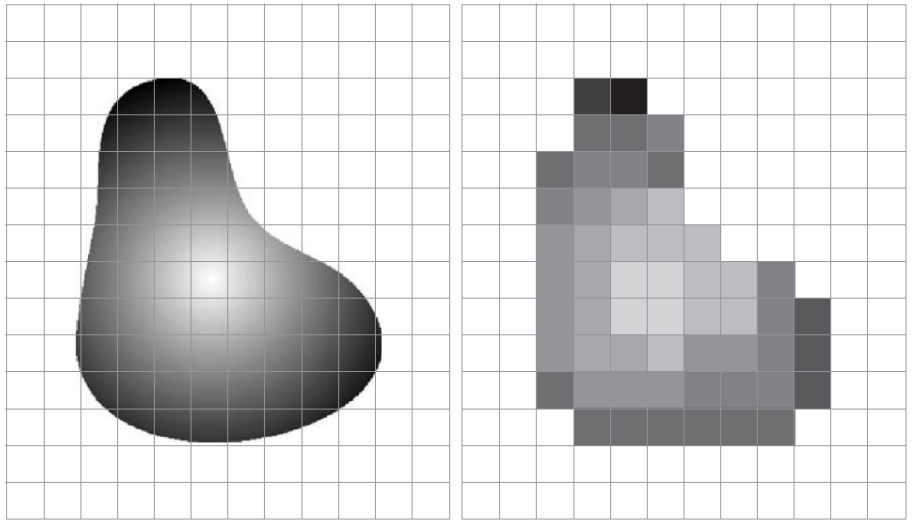








digitizing the coordinate values: pixels



digitizing the amplitude values: discrete intensity levels

Discrete Intensity Levels

image data: 2D grid (matrix) of pixels with different intensity values

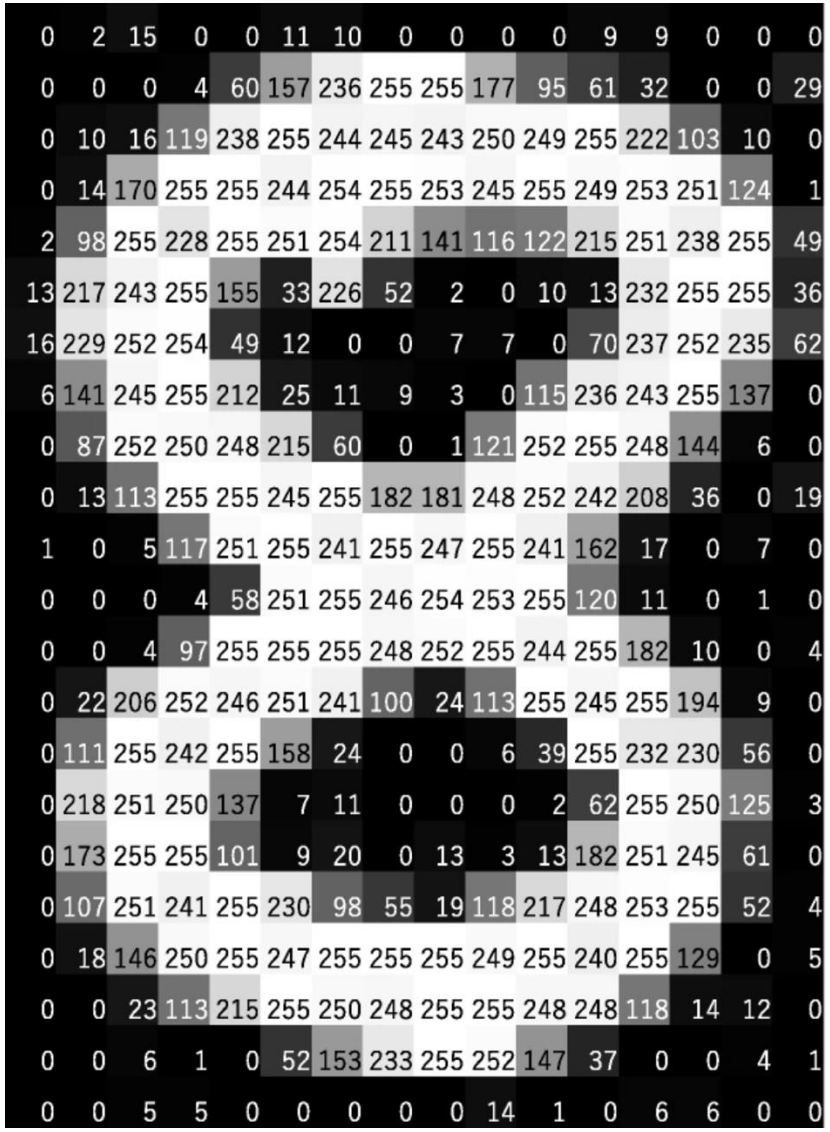
intensity resolution:

common to use one byte (2^8) to express possible intensity values (8-bit image)

→ 0 = black, 255 = white

storage requirement for, e.g., 1024×1024 8-bit image (grayscale): ~1 MB

(× 3 for RGB color images)



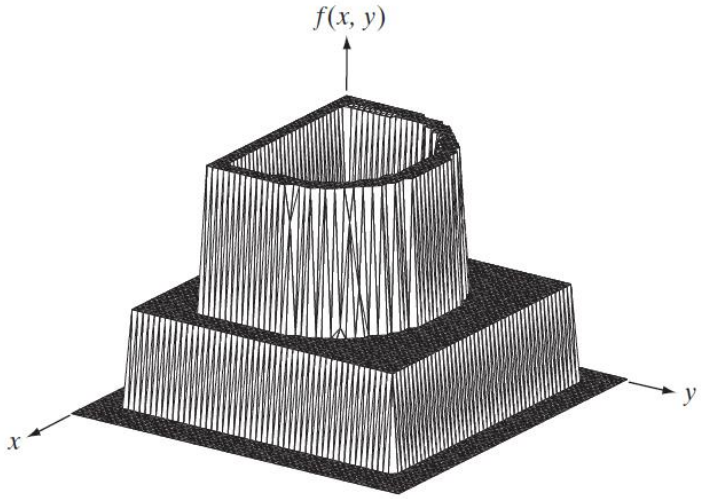
source

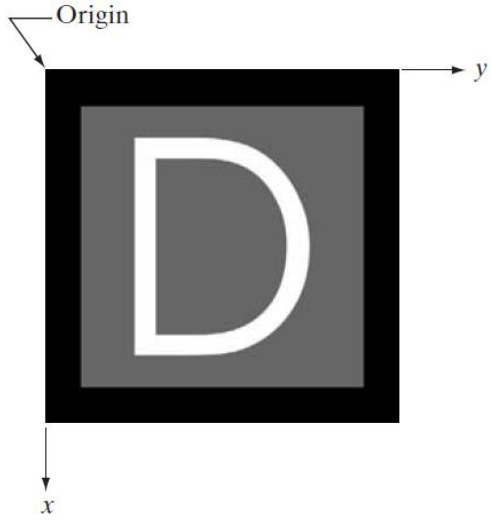
Image as Two-Dimensional Function

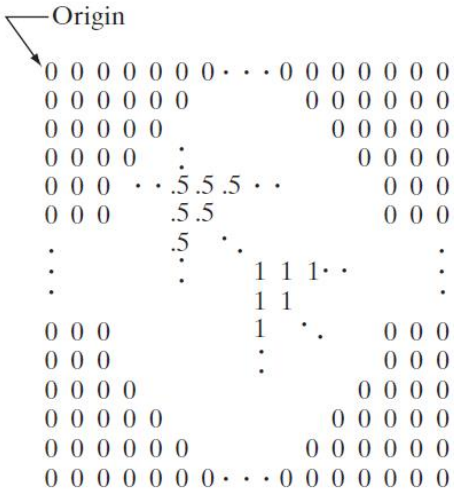
image can be seen as function $f(x, y)$: intensity at position (x, y)

digital image: discrete (sampled and quantized) version of this function

$$f(x, y) = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0,N-1) \\ f(1,0) & f(1,1) & \cdots & f(1,N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \cdots & f(M-1,N-1) \end{bmatrix}$$







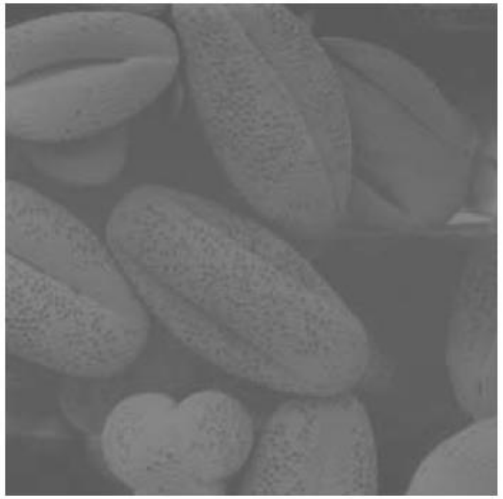
Contrast

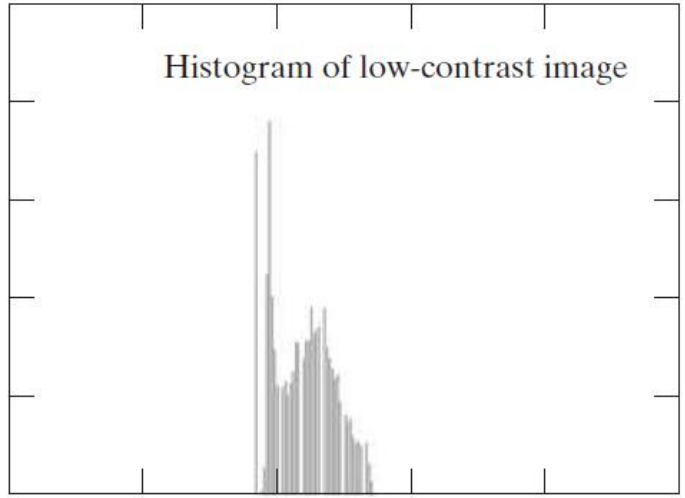
range of intensity levels:

- maximum measurable intensity determined by saturation
- minimum detectable intensity determined by noise

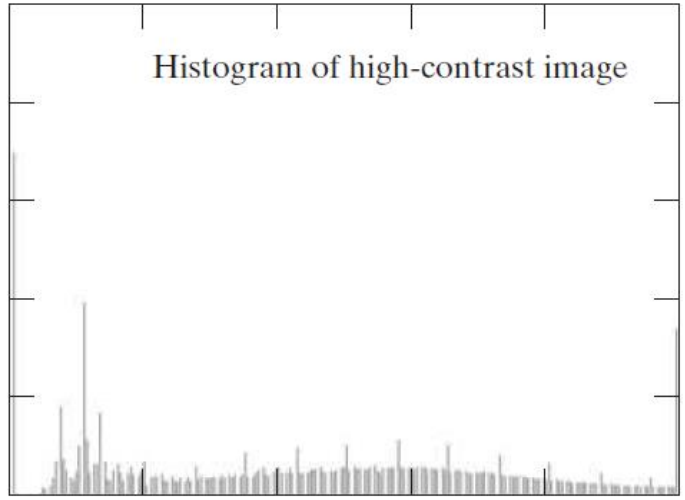
contrast: difference in intensity between highest and lowest intensity levels in an image

intensity histograms:







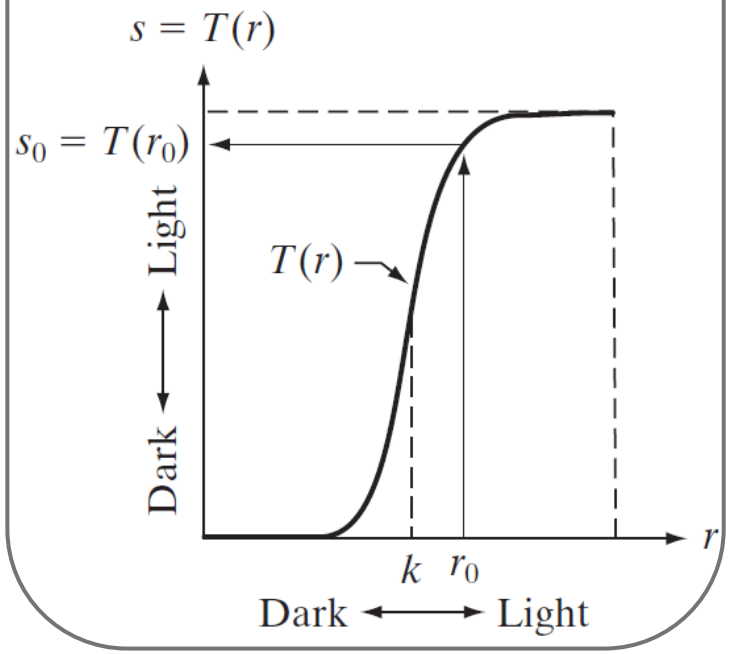


lighten or darken image simply by adding or subtracting a fixed value from the pixel intensities

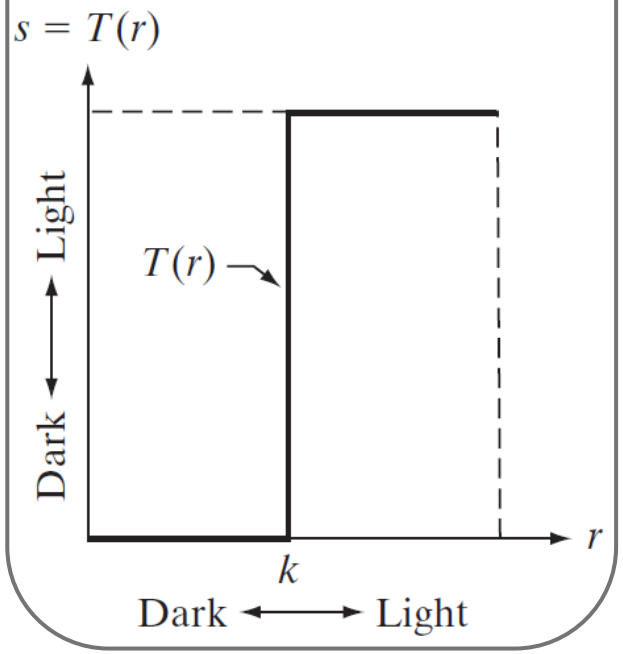
Intensity Transformations

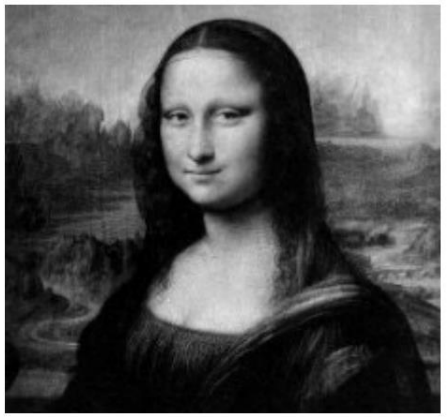
e.g., image enhancement through contrast manipulation

contrast stretching (expanding range of intensity levels):



thresholding (differentiation of foreground and background regions):



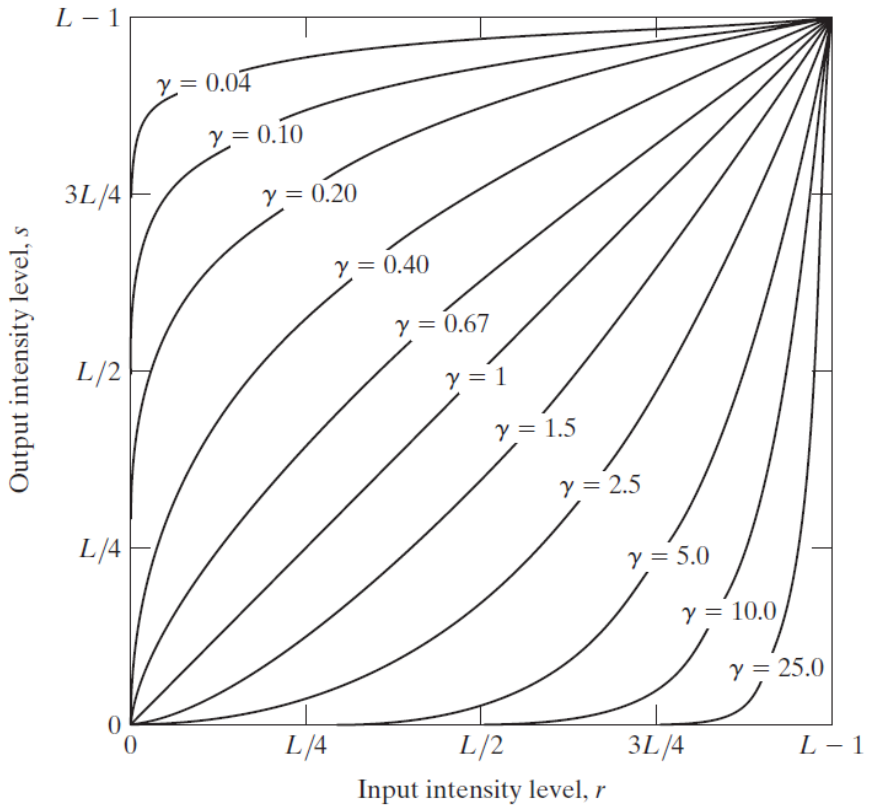




Power-Law Transformations

simultaneous spreading and compression of intensity levels

e.g., gamma correction:



$$s = r^{\gamma}$$

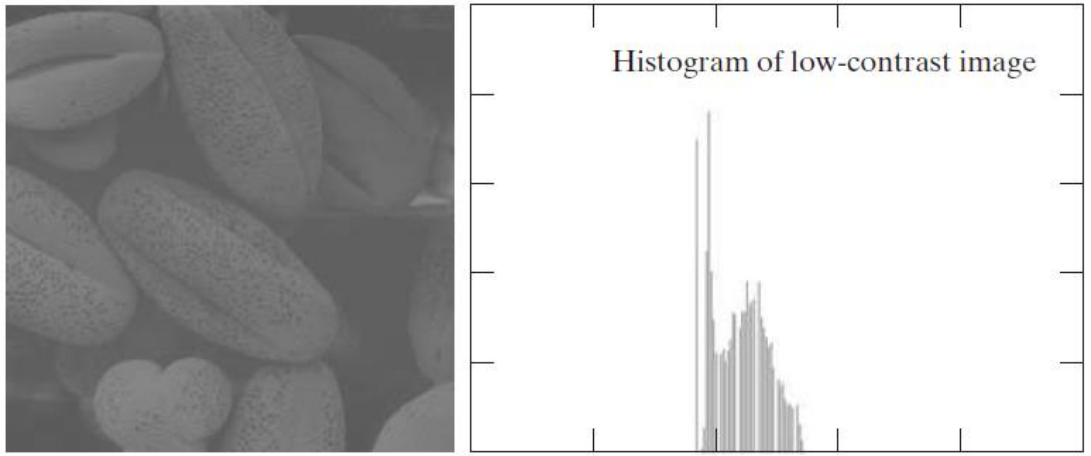


Histogram Equalization

automatic contrast stretching

global transformation function: based on intensity distribution of entire image

local histogram equalization by restricting transformation to pixel neighborhoods



after histogram equalization:

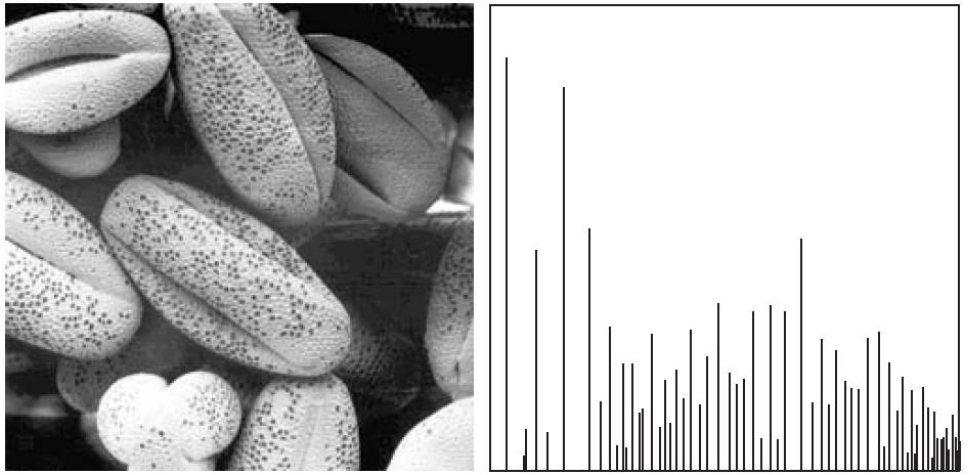


image size: number of pixels

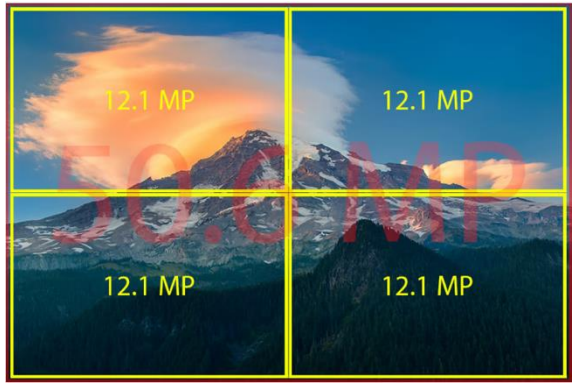
Spatial Resolution

pixels per unit distance: e.g., pixels per inch (ppi)

→ determines print size of digital image with given pixel count



same resolutions



look at images in screen resolution (property of digital screen)

e.g., iPhone 16 Pro Max: 2868×1320 pixels, 6.9 inches diagonal, 460 ppi

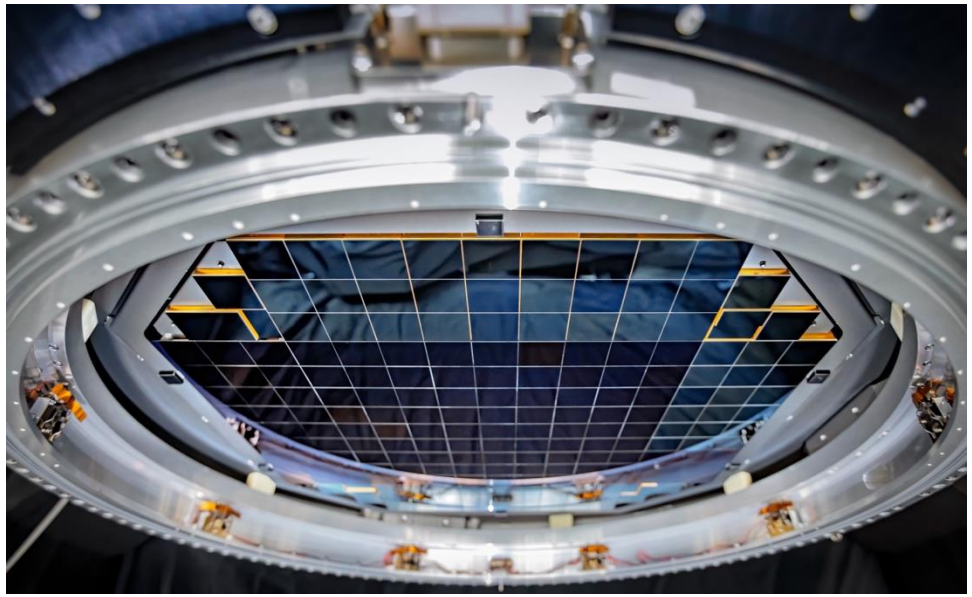
→ either fix number of pixels and calculate figure size, or fix figure size and resample number of pixels (interpolation)

term for printers: dots per inch (dpi) → property of printer

Example from Astronomy

Vera C. Rubin Observatory

first light in 2025

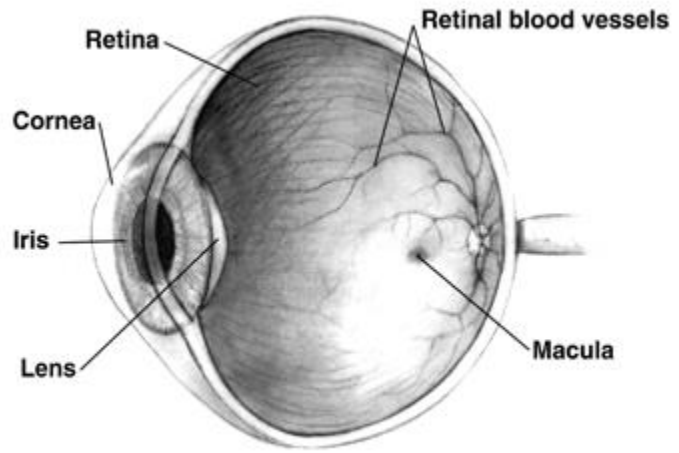


largest digital camera ever made: 3.2-gigapixel CCD



Color

Colors in the Human Eye

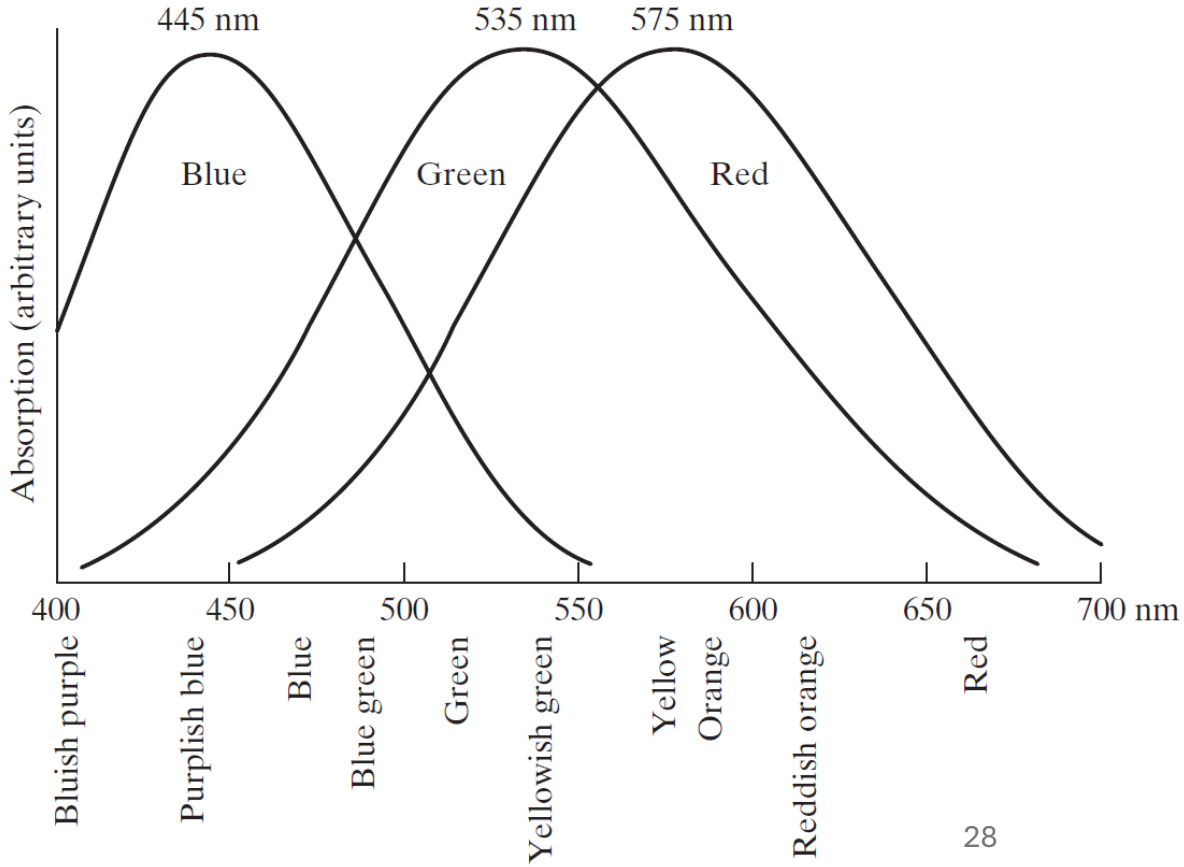


retina: image sensor of human eye

two kinds of light receptors on surface of retina: rods (monochromatic vision) and cones (color-sensitve cells)

three types of cones: red, green, blue

→ combinations of red, green, and blue create perceived colors



Primary Colors

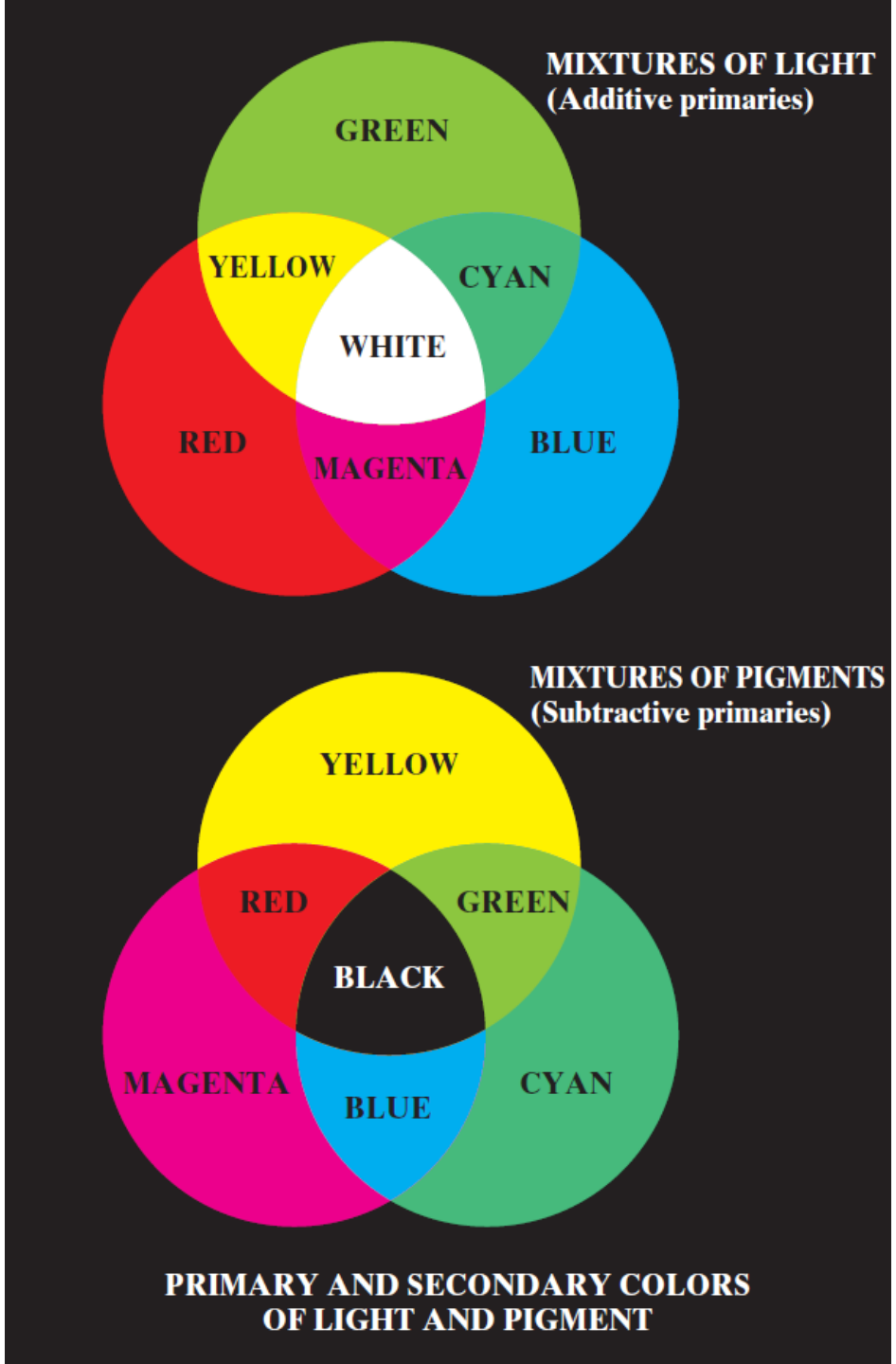
standardization:

use of specific wavelengths for three primary colors red, green, and blue

→ can reproduce (almost) all visible colors by mixing with different intensities

for printing:

primary colors of pigments each absorb one primary color of light

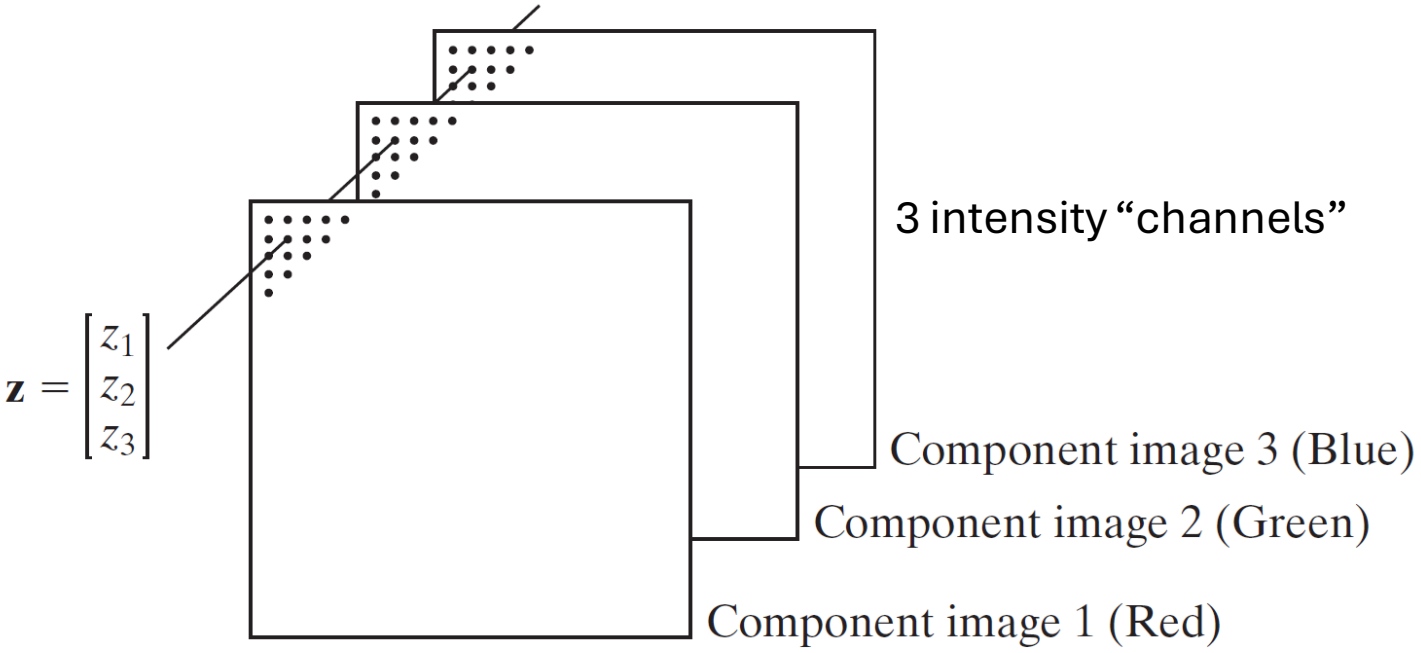


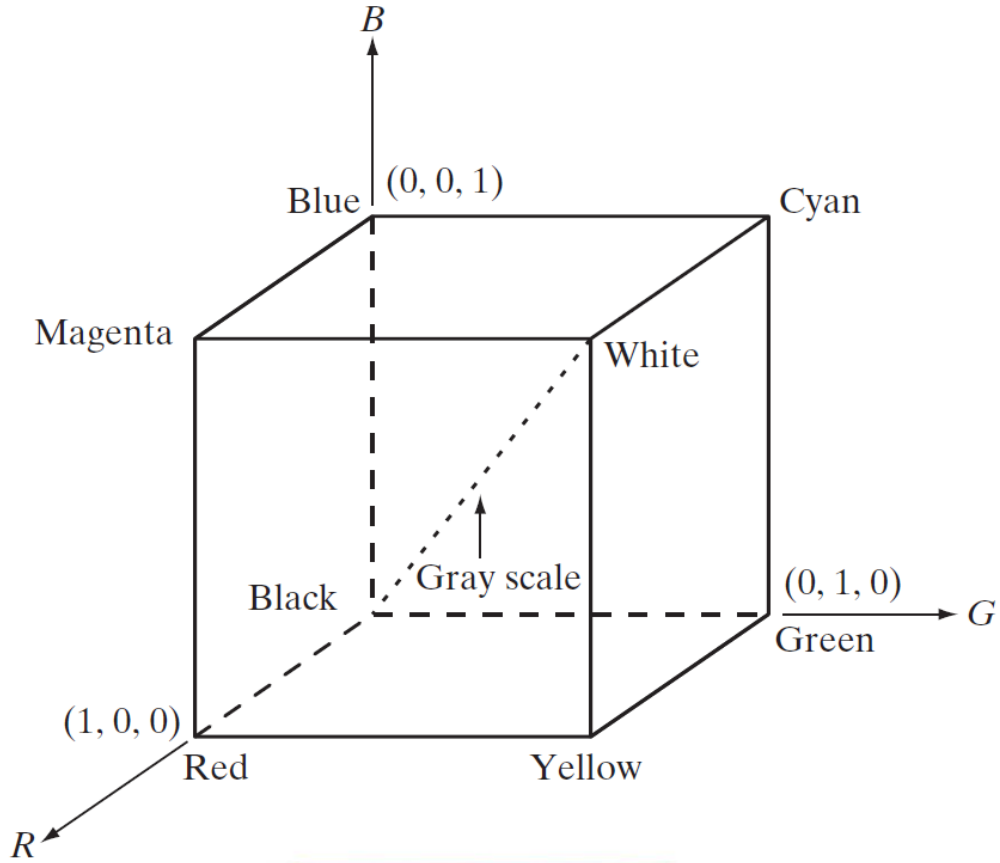
RGB Color Model

color model for monitors and cameras

(other color models: CMYK, HSV, …)

color pixels:





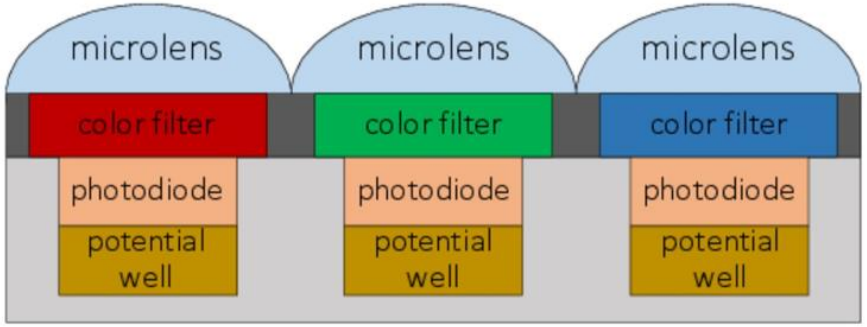


Color Sensing

single image sensor with one of three color filters in front of each pixel

estimation of missing color values for each pixel by interpolation from neighboring pixels

→ full-color image

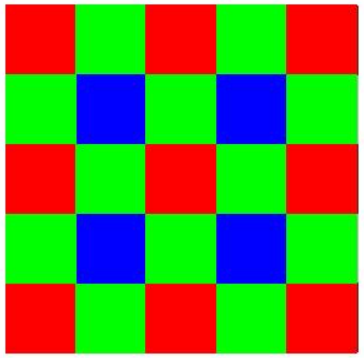


G	R	G	R
B	G	B	G
G	R	G	R
B	G	B	G

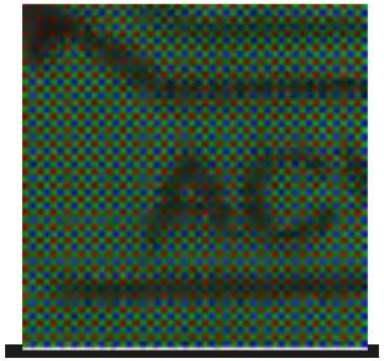
Bayer RGB Pattern

rGb	Rgb	rGb	Rgb
rgB	rGb	rgB	rGb
rGb	Rgb	rGb	Rgb
rgB	rGb	rgB	rGb

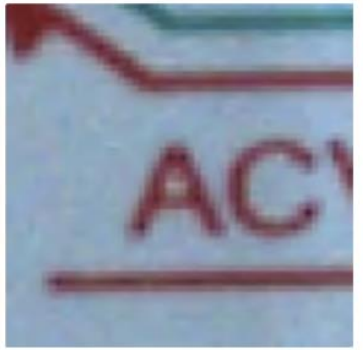
Interpolated Pixels



Bayer Pattern
(Color Filter Mosaic)



Raw Image



Interpolated Image

Image Compression

Need for Compression

crucial for storage and transmission

example HD TV:

image resolutions of $1920 \times 1080 \times 24$ bits

digital video as sequence of full-color still images (frames) with 30 frames per second

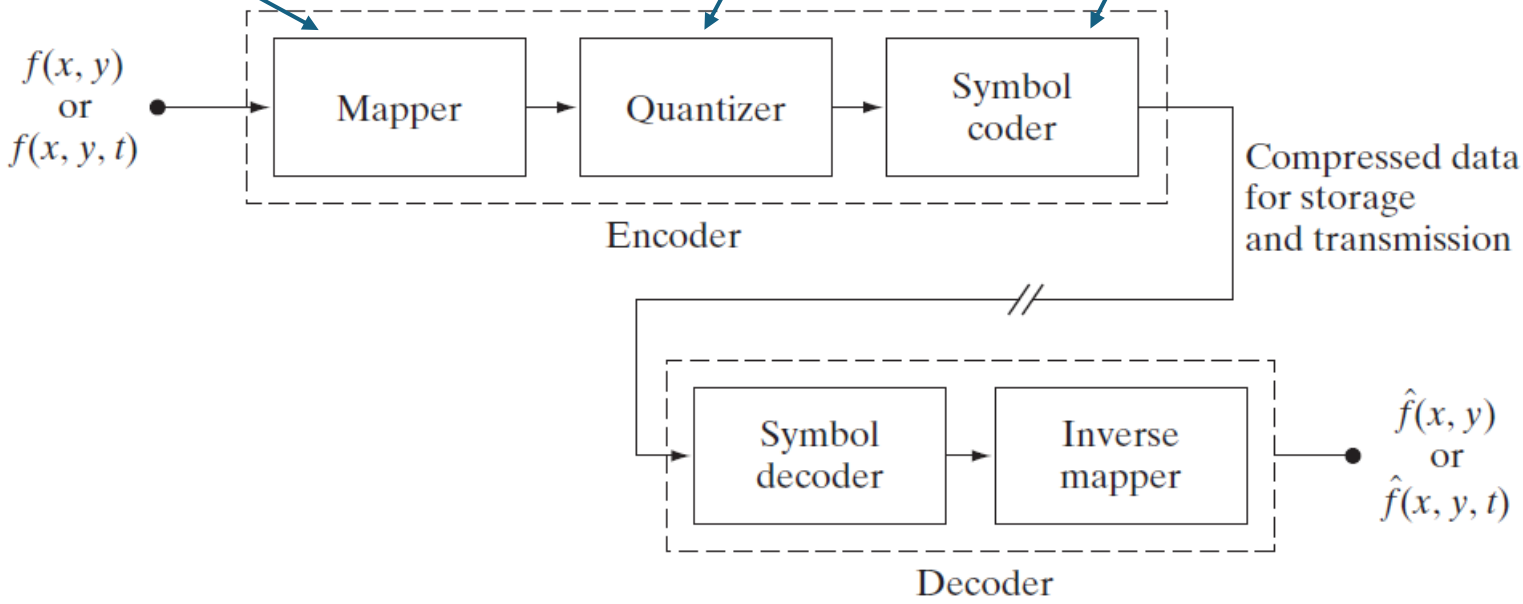
→ 2-hours video 1.34 TB

Compression & Decompression

reduce spatial and temporal redundancy: exploit correlations between pixels

remove “irrelevant visual” information → irreversible process (lossy compression)

minimizing coding redundancy: shortest code words assigned to most frequent intensity values (variable-length code)



JPEG, GIF, PNG, …

Example: Block Transform Coding

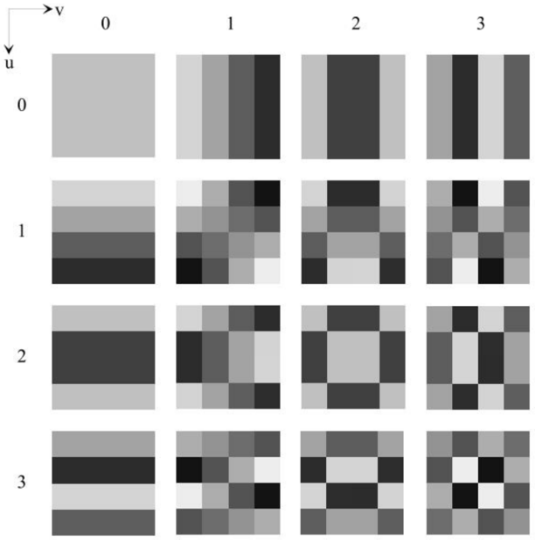
decorrelate pixels of each sub-image by means of discrete 2D transform

- e.g., discrete cosine transform (DCT) or wavelet transform
- DCT kind of approximated principal component analysis (PCA)

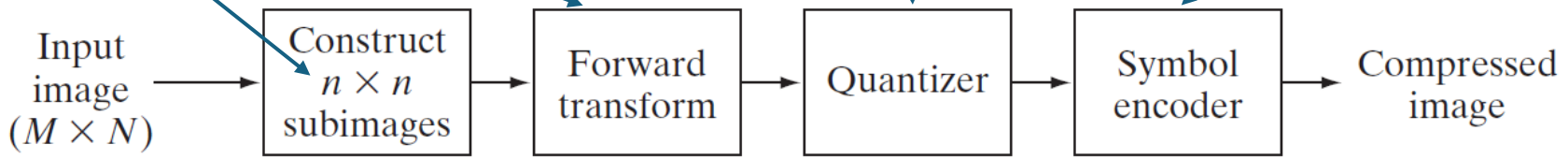
drop or coarsely quantize least important transform coefficients

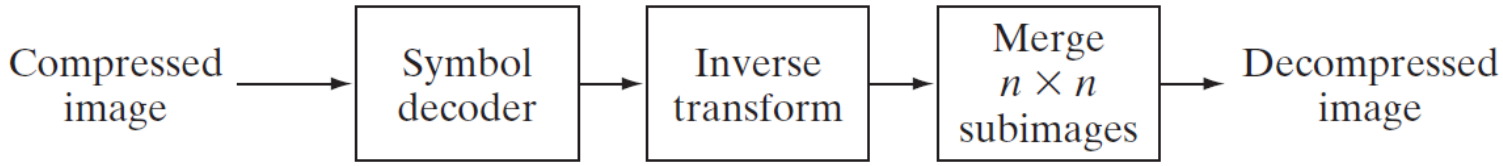
encode quantized coefficients (e.g., Huffman coding)

DCT basis functions:



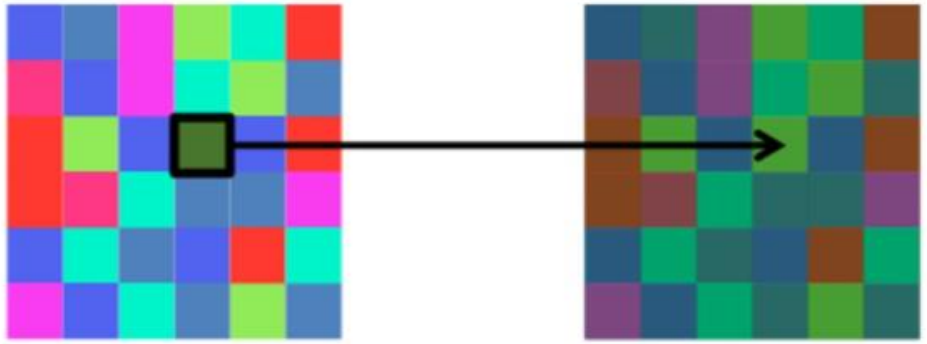
often 8×8





Spatial Filtering

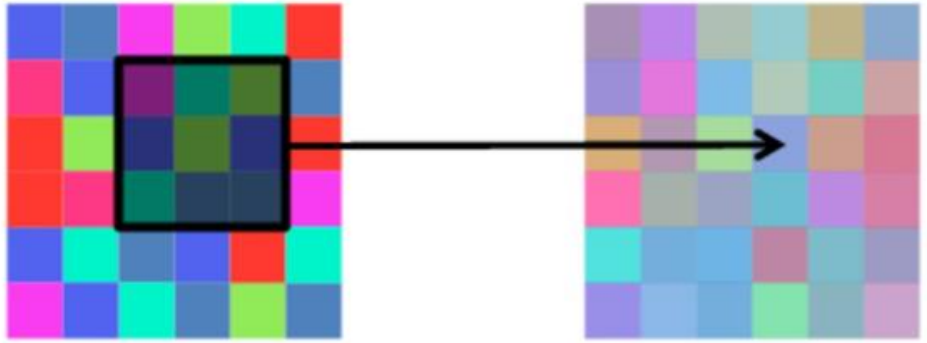
Single-Pixel/Point Operation



point processing

intensity transformations

Neighborhood Operation



“filtering”

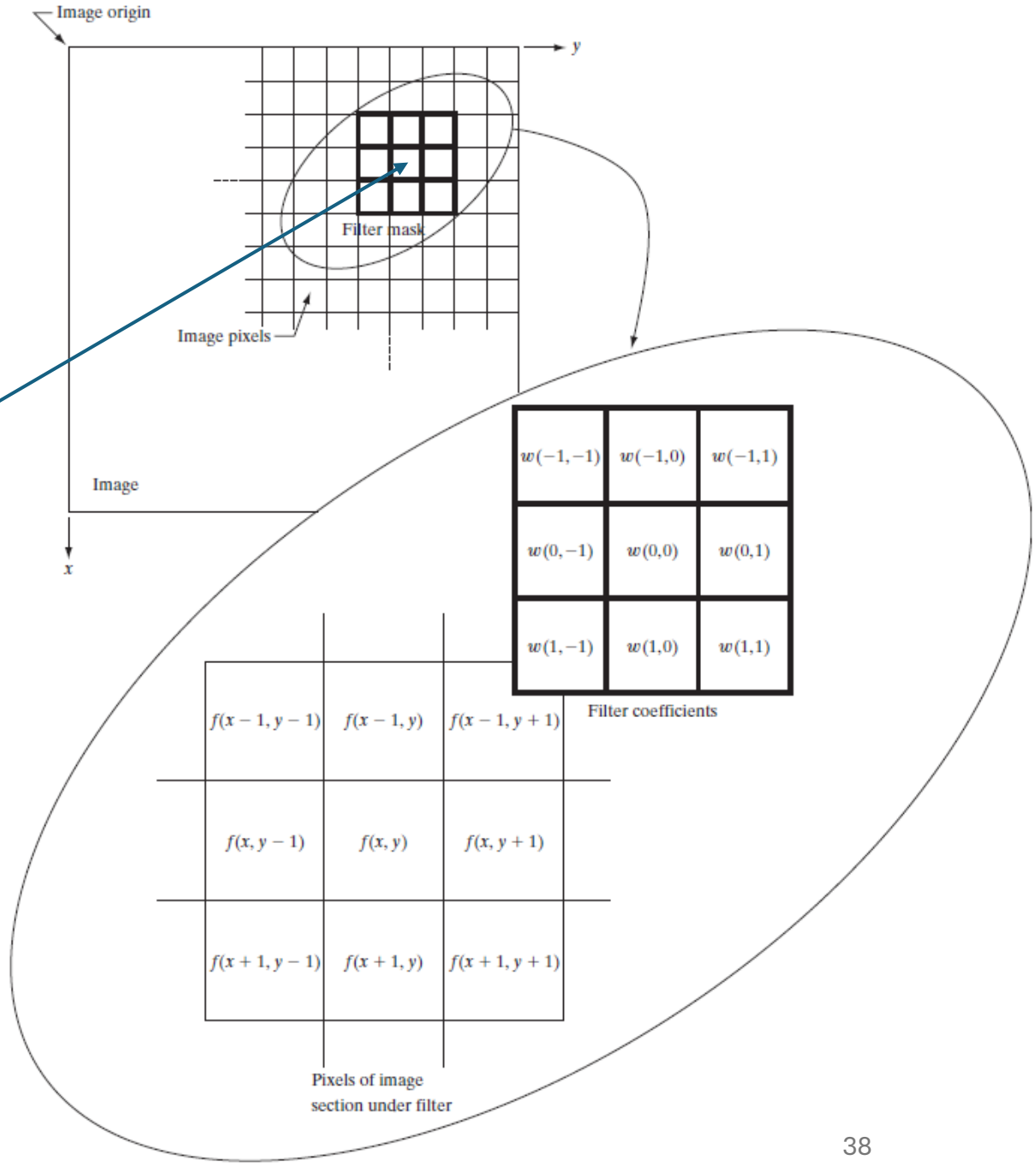
Spatial Filtering

linear shift-invariant operators:

- replace each pixel's intensity by weighted sum of its neighborhood
- weights determined by a kernel (aka mask or filter)
- same kernel **shifted** to all pixels

for example, filter of size 3×3:

$$g(x,y) = \sum_{s=-1}^{1} \sum_{t=-1}^{1} w(s,t) f(x+s, y+t)$$



We set the filter coefficients manually here. Later, we will use a CNN for this.

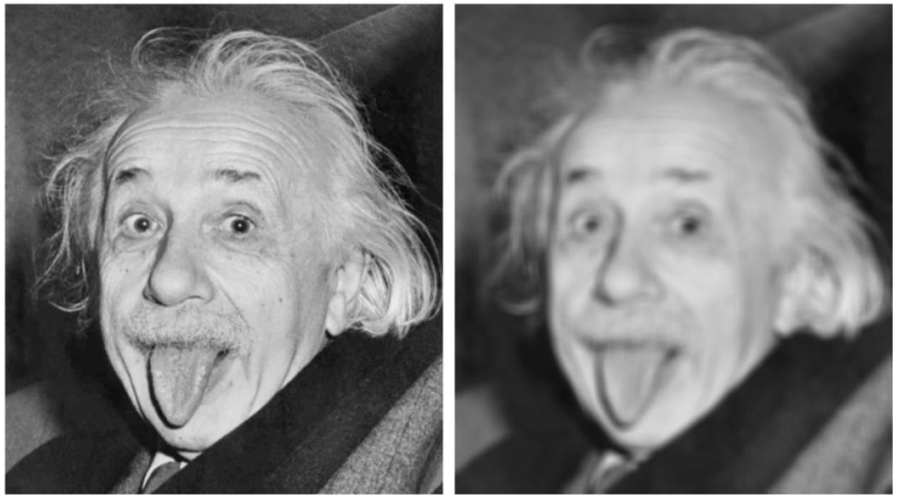
Example: Box Filter

$$w = \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \qquad m \times m: w = \frac{1}{m^2}\begin{bmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{bmatrix}$$

the larger the kernel, the heavier the smoothing effect

results in local (moving) averages

→ blurring (smoothing) effect:



- without normalization resulting image would be saturated
- the larger the filter size, the heavier the blurring effect
- filters of even size possible as well, but odd-sized ones easier to deal with

Definition: Correlation vs Convolution

using our example with filter of size 3×3

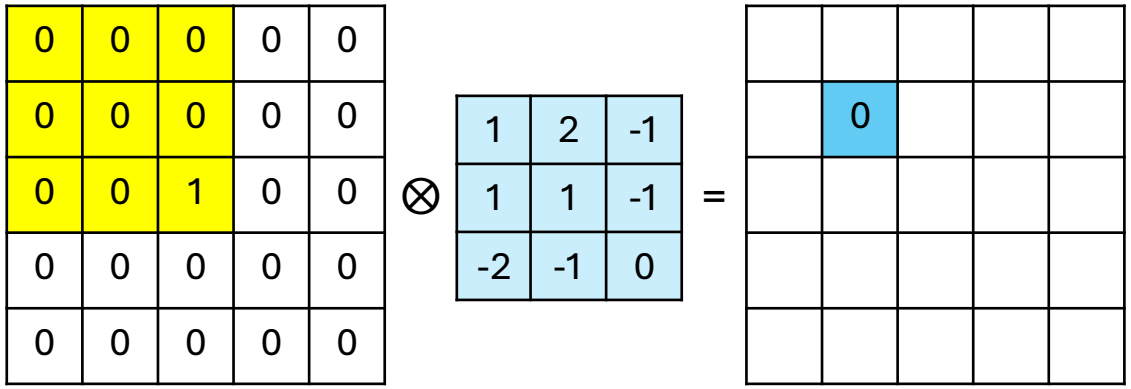
correlation: $g(x,y) = \sum_{s=-1}^{1}\sum_{t=-1}^{1} w(s,t) f(x+s, y+t)$

convolution: $g(x,y) = \sum_{s=-1}^{1}\sum_{t=-1}^{1} w(s,t) f(x-s, y-t)$

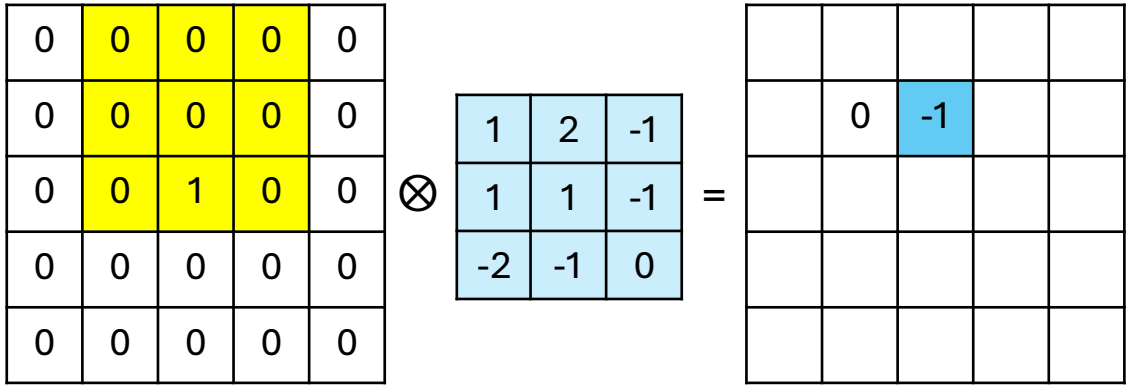
corresponds to flip of filter

- convolution of a kernel with a discrete unit impulse yields copy of the kernel at location of the impulse (and impulse filter yields image copy)
- corresponding correlation instead yields rotated version of the kernel (or rotated image)

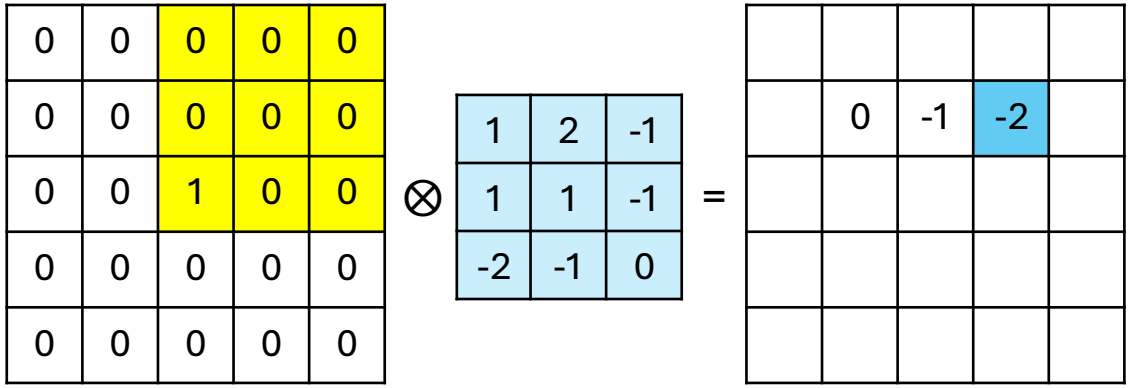
step 1:



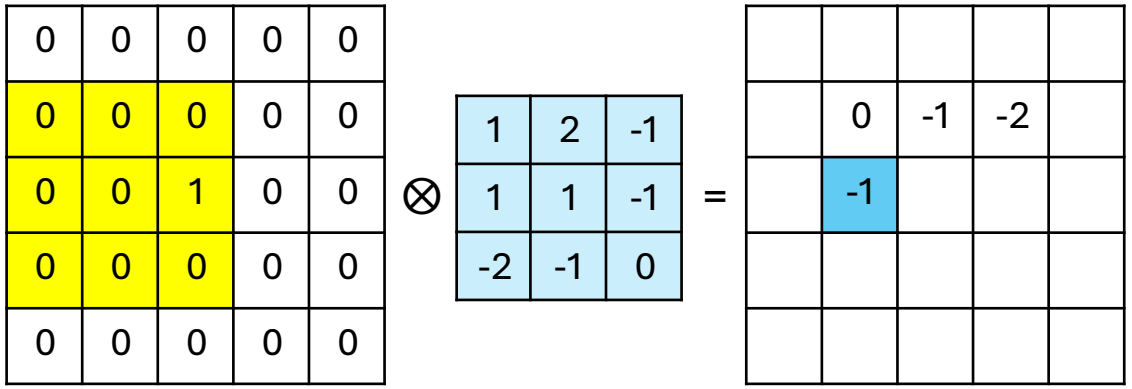
step 2:



step 3:

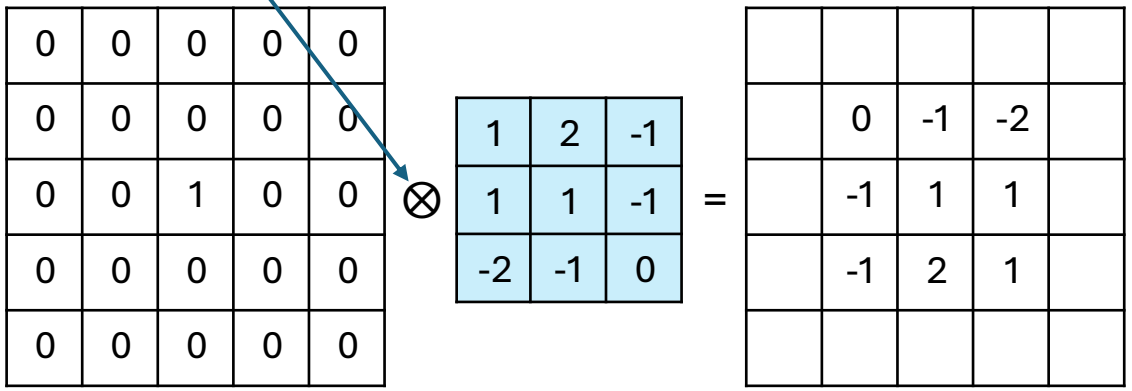


step 4:

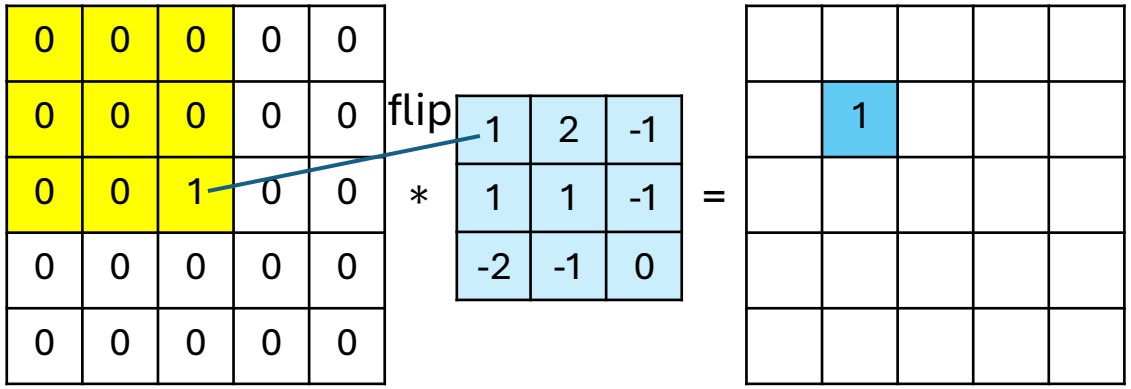


correlation

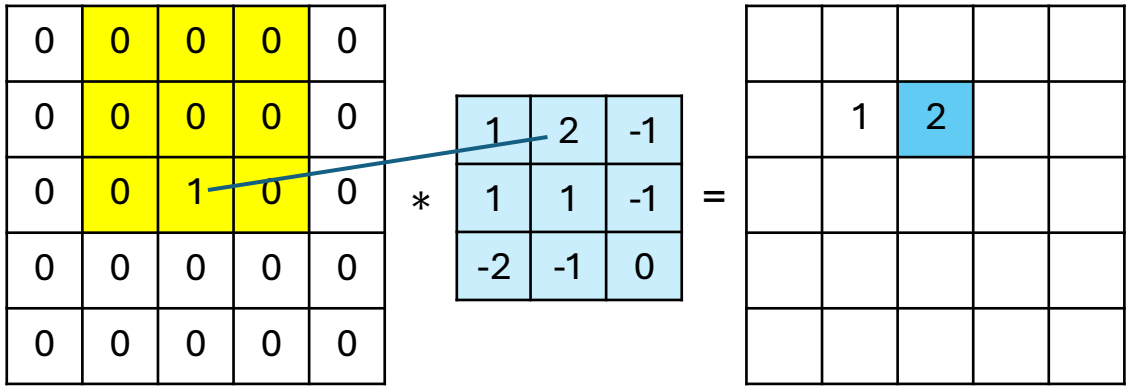
result:



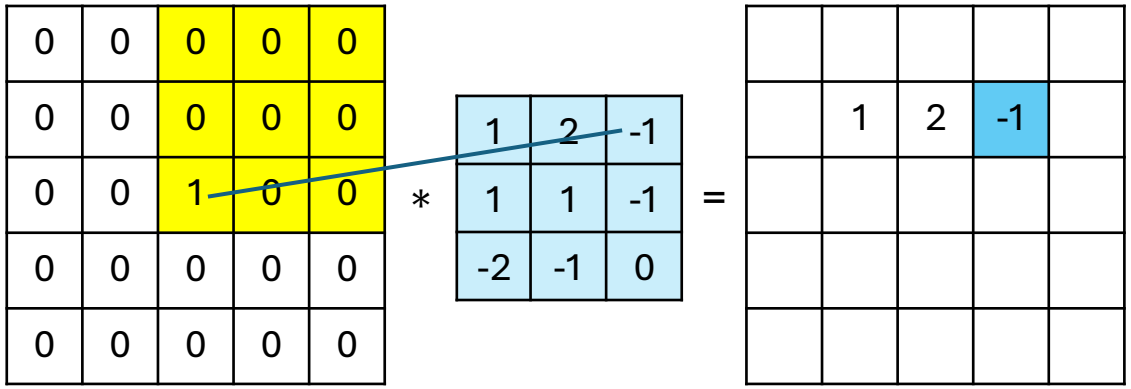
step 1:



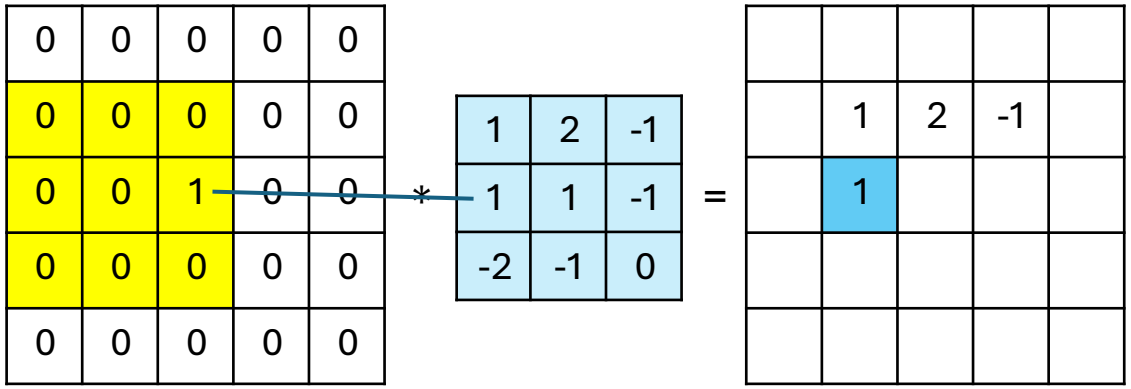
step 2:



step 3:

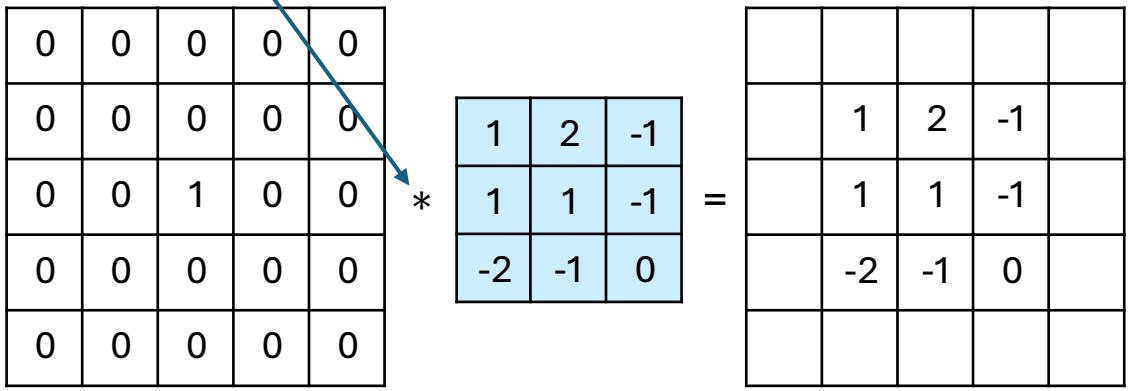


step 4:



convolution

result:



Padding

On last slides, we did not let the filter extend beyond image borders.

→ boundary effects: resulting image smaller than original

zero-padding to avoid this:

0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

∗

1	2	-1
1	1	-1
-2	-1	0

=

0	0	0	0	0
0	1	2	-1	0
0	1	1	-1	0
0	-2	-1	0	0
0	0	0	0	0

zero-padding darkens the image edges

other padding options:

- set to constant border value
- repeat edge values
- …

can also be more zero rows/columns (depending on image and filter size)

Other Linear Smoothing Filters

when constructing filters (not only smoothing filters, but in general), look for separable ones:

allow to compute 2D convolution as two subsequent 1D convolutions (horizontal and vertical)

for image of size $M \times M$ and kernel of size $N \times N$:

- $M^2 \cdot N^2$ operations for non-separable filter
- $M^2 \cdot 2 \cdot N$ operations for separable filter

→ significant speedup possible

sampled from 2D Gaussian function (larger σ requires larger kernel)

bilinear filter (tent):

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Gaussian filter:

$$\frac{1}{256}\begin{bmatrix} 1 & 4 & 6 & 4 & 1 \\ 4 & 16 & 24 & 16 & 4 \\ 6 & 24 & 36 & 24 & 6 \\ 4 & 16 & 24 & 16 & 4 \\ 1 & 4 & 6 & 4 & 1 \end{bmatrix}$$

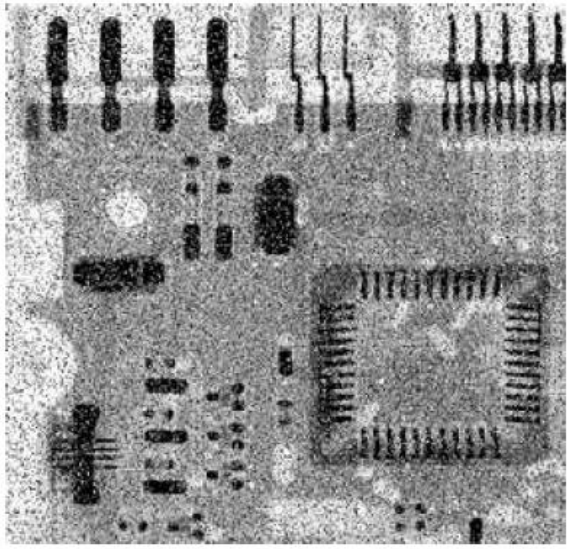
$$\frac{1}{4}\begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \qquad \frac{1}{16}\begin{bmatrix} 1 & 4 & 6 & 4 & 1 \end{bmatrix}$$

corresponding horizontal 1D kernels

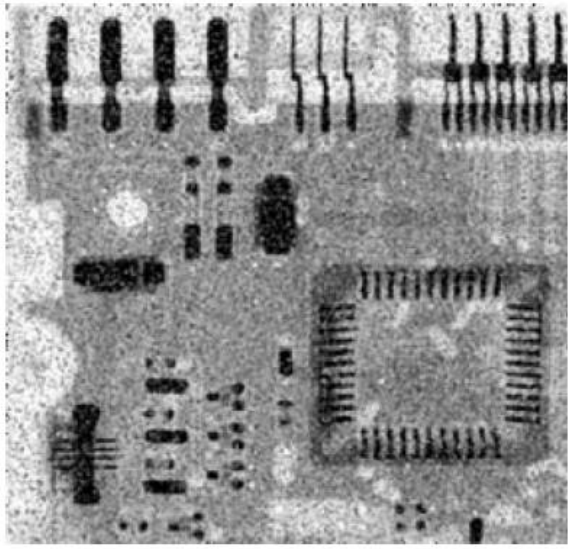
Median Filter

- same kind of neighborhood-shifting over image
- but instead of weighted average, use median of intensity values in pixel neighborhood (no filter coefficients) → non-linear filter
- very effective to reduce outliers, e.g., shot or salt-and-pepper noise

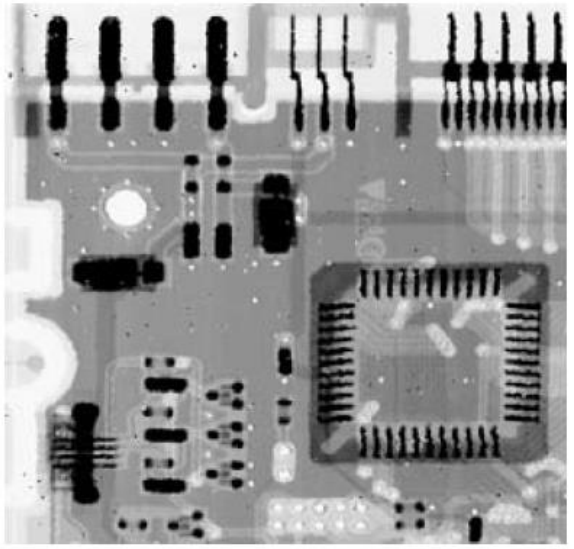
original:



average-filtered (3×3):



median-filtered (3×3):



playground

Sharpening Filters

goal of smoothing filters:

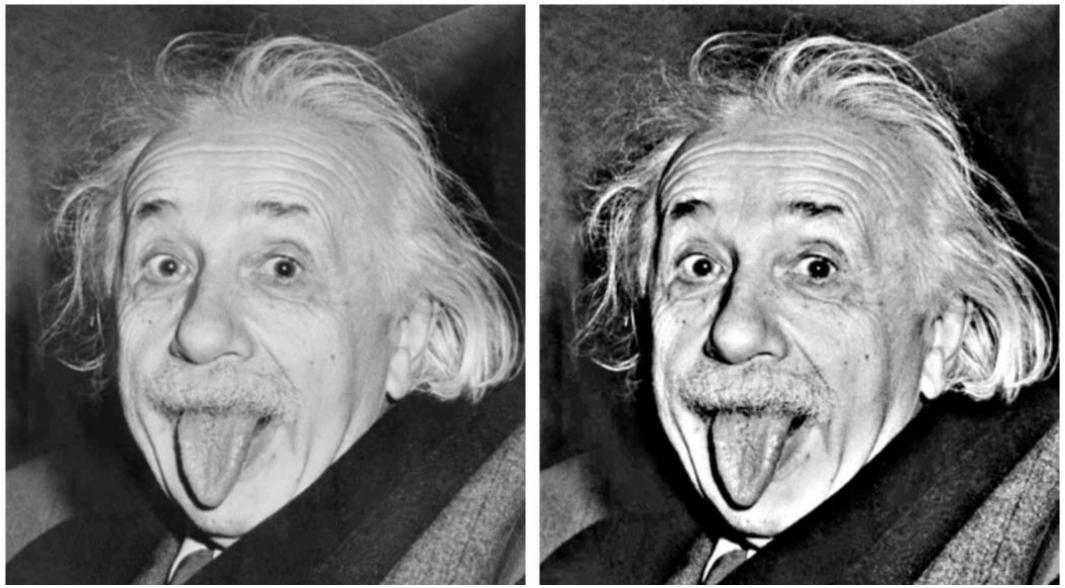
removal of small details, noise reduction

goal of sharpening filters:

highlight intensity transitions (e.g., edges)

averaging analogous to integration

→ sharpening by differentiation



First- and Second-Order Derivatives

partial derivatives in discrete form:

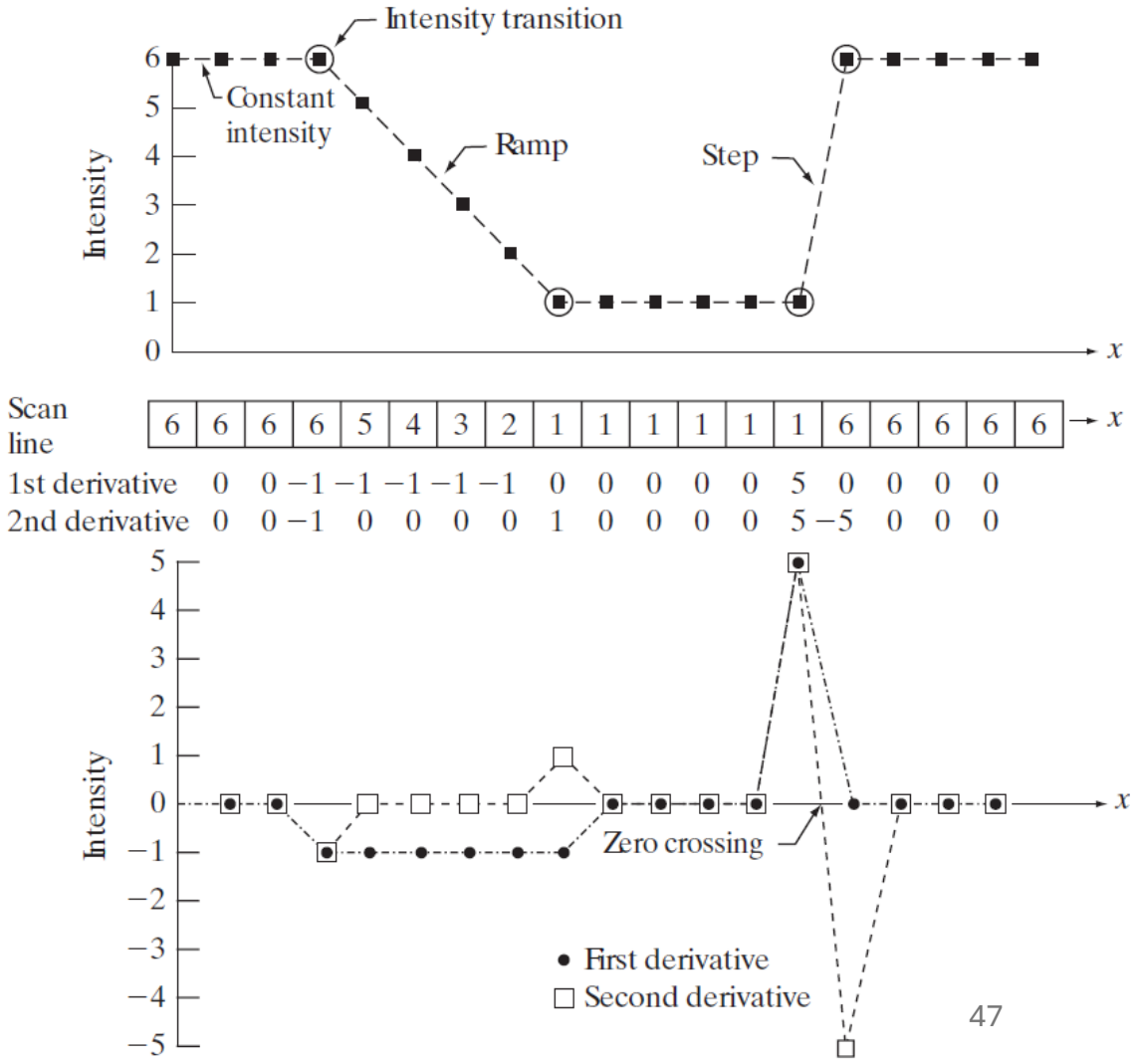
$$\frac{\partial f}{\partial x} = f(x+1, y) - f(x, y)$$

$$\frac{\partial f}{\partial y} = f(x, y+1) - f(x, y)$$

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

1D visualization:



Approximated Image Gradient

alternative formulation (1D): $\frac{\partial f}{\partial x} = \frac{f(x+1) - f(x-1)}{2}$

→ 1D derivative filter:

1	0	-1

Sobel filters:

smoothing

1
2
1

$*$ horizontal

-1	0	1

$=$

1	0	-1
2	0	-2
1	0	-1

$\cdot \frac{1}{8}$

construct separable filters:

(convolution kernels)

vertical

-1
0
1

$*$ smoothing

1	2	1

$=$

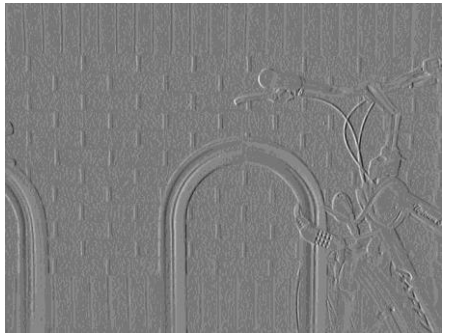
1	2	1
0	0	0
-1	-2	-1

$\cdot \frac{1}{8}$

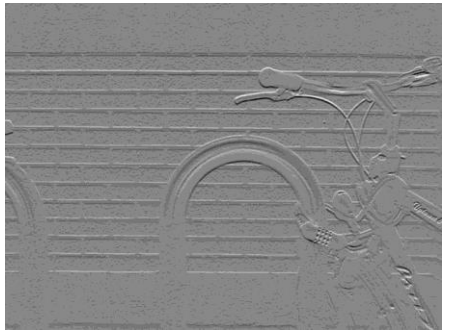
→ gradient with smoothing

convolved with an image:





$\frac{\partial f}{\partial x}$



$\frac{\partial f}{\partial y}$

Magnitude of Image Gradient

magnitude/amplitude of gradient:

$$\|\nabla f\| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$

magnitude is isotropic

for sharpening:
add this to original



Different Gradient Filters

Gradient	Roberts	Prewitt	Sobel (3x3)	Sobel (5x5)
$\frac{\partial I}{\partial x}$	0 1 -1 0	-1 0 1 -1 0 1 -1 0 1	-1 0 1 -2 0 2 -1 0 1	-1 -2 0 2 1 -2 -3 0 3 2 -3 -5 0 5 3 -2 -3 0 3 2 -1 -2 0 2 1
$\frac{\partial I}{\partial y}$	1 0 0 -1	1 1 1 0 0 0 -1 -1 -1	1 2 1 0 0 0 -1 -2 -1	1 2 3 2 1 2 3 5 3 2 0 0 0 0 0 -2 -3 -5 -3 -2 -1 -2 -3 -2 -1

Good Localization
Noise Sensitive
Poor Detection

→

Poor Localization
Less Noise Sensitive
Good Detection

the larger ones add smoothing components (Sobel: convolution with bilinear filter)

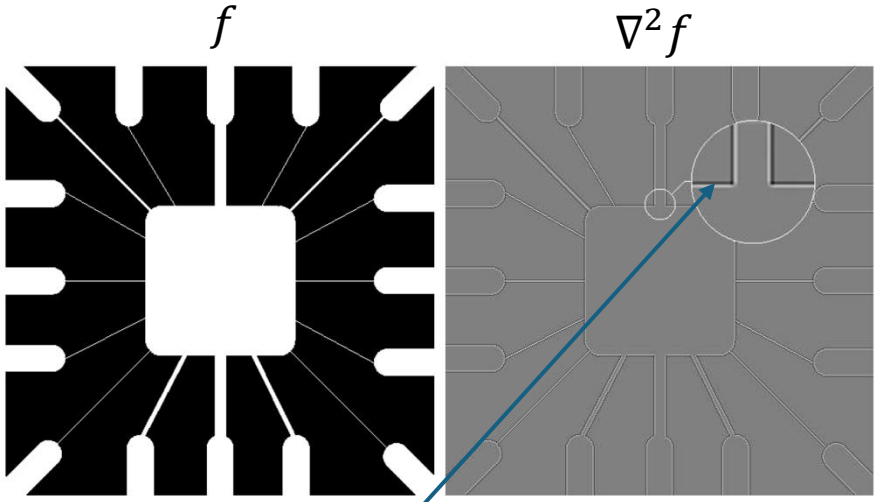
Alternative: Laplacian

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

filter mask:

0	1	0
1	-4	1
0	1	0

$$= f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$



scaled to visualize negative values

zero-crossings → edges

isotropic for rotations in increments of 90°

to yield isotropic results for rotations in increments of 45°:

1	1	1
1	-8	1
1	1	1

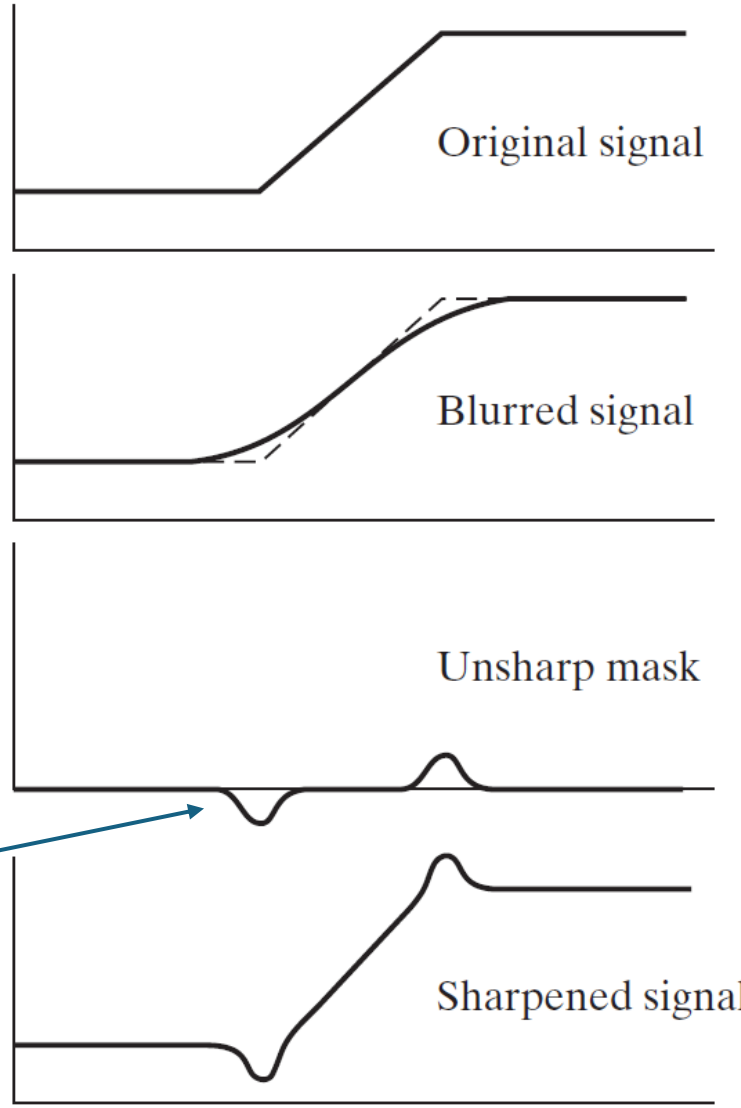
again, add to (or rather subtract from when using this filter definition here) original to get sharpened image:

$$g(x, y) = f(x, y) - \nabla^2 f(x, y)$$

Another Alternative: Unsharp Masking

use of smoothing filters for sharpening

1. smoothing of original image (e.g.,by Gaussian filter)
2. subtract the blurred image from the original → unsharp mask (similar to second-order derivative)
3. add unsharp mask to original image → sharpened image



similar to second-order derivative


original

−


smoothed (3x3)

=


Detailed


original

+


Detailed

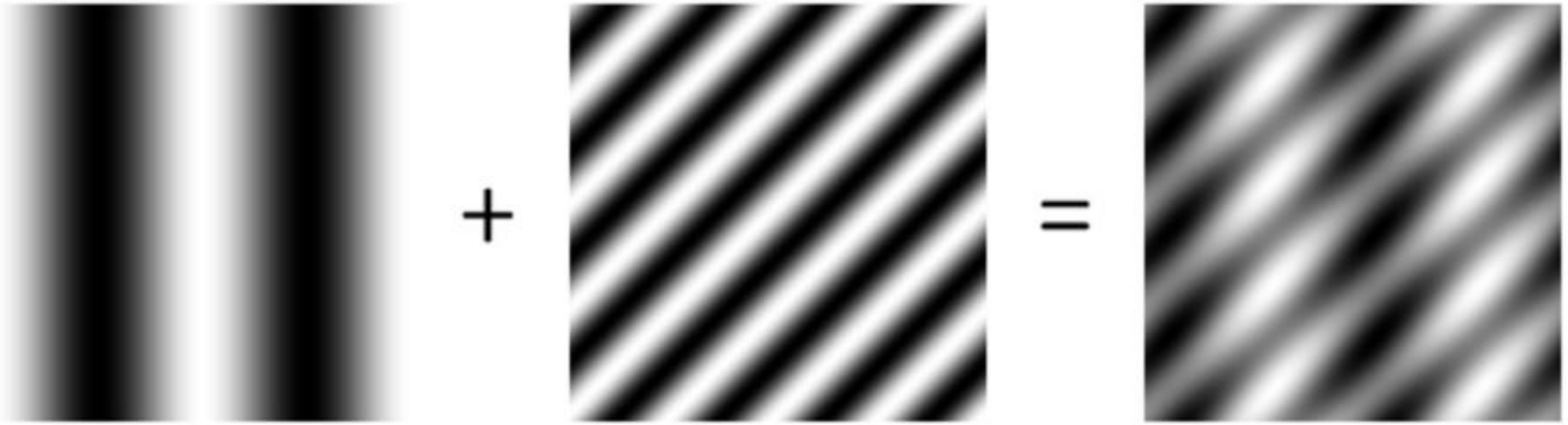
=


Sharpened

Fourier Transform

Fourier analysis allows to determine frequency characteristics of images or filters

image



frequency content



Frequency Representation of Spatial Function

response of a spatial function (filter) $h(x)$ to a sinusoid with frequency ω:

$$h(x) * e^{i\omega x} = Ae^{i(\omega x+\phi)}$$

angular frequency $\omega = 2\pi f$
notation: f sometimes u or μ

Fourier transform: tabulation of magnitude A and phase ϕ response **at each frequency**

$$H(\omega) = \mathcal{F}\{h(x)\} = Ae^{i\phi}$$

complex: needs to capture both amplitude and phase

Fourier transform pair: $h(x) \overset{\mathcal{F}}{\leftrightarrow} H(\omega)$

frequency domain

Fourier Transform Formulas

for continuous signals:

$$H(\omega) = \int_{-\infty}^{\infty} h(x) e^{-i\omega x} dx$$

inverse Fourier transforms:

$$h(x) = \int_{-\infty}^{\infty} H(\omega) e^{i\omega x} d\omega$$

Discrete Fourier Transform (DFT):

$$H(k) = \sum_{x=0}^{N-1} h(x) e^{-i\frac{2\pi k x}{N}}$$

$$h(x) = \frac{1}{N} \sum_{k=0}^{N-1} H(k) e^{i\frac{2\pi k x}{N}}$$

unique only in range $k \in \left[-\frac{N}{2}, \frac{N}{2}\right]$

2D Fourier Transforms

$$H(\omega_x, \omega_y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) e^{-i(\omega_x x + \omega_y y)} dx dy$$

$$H(k_x, k_y) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} h(x, y) e^{-i2\pi(k_x x/M + k_y y/N)}$$

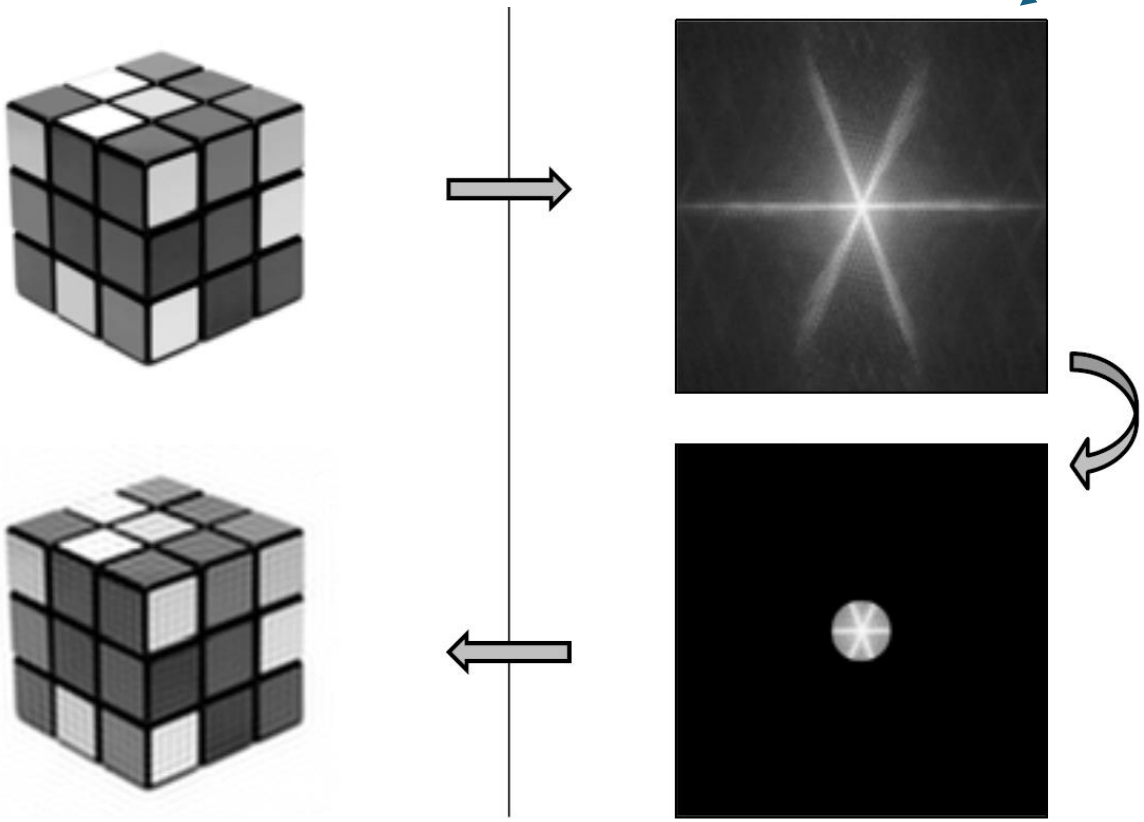


e.g., digital images (M and N as width and height in pixels)

Filtering in Frequency Domain

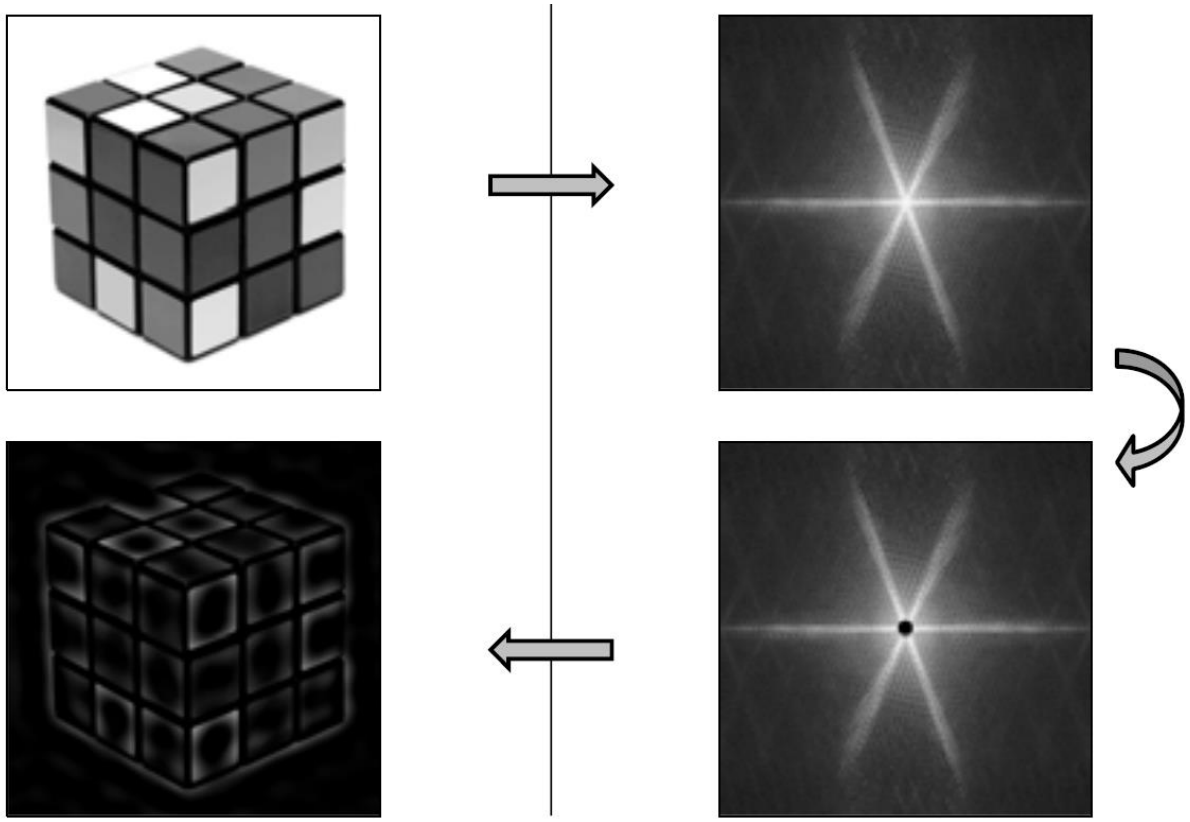
low-pass filtering:

magnitude



→ smoothing

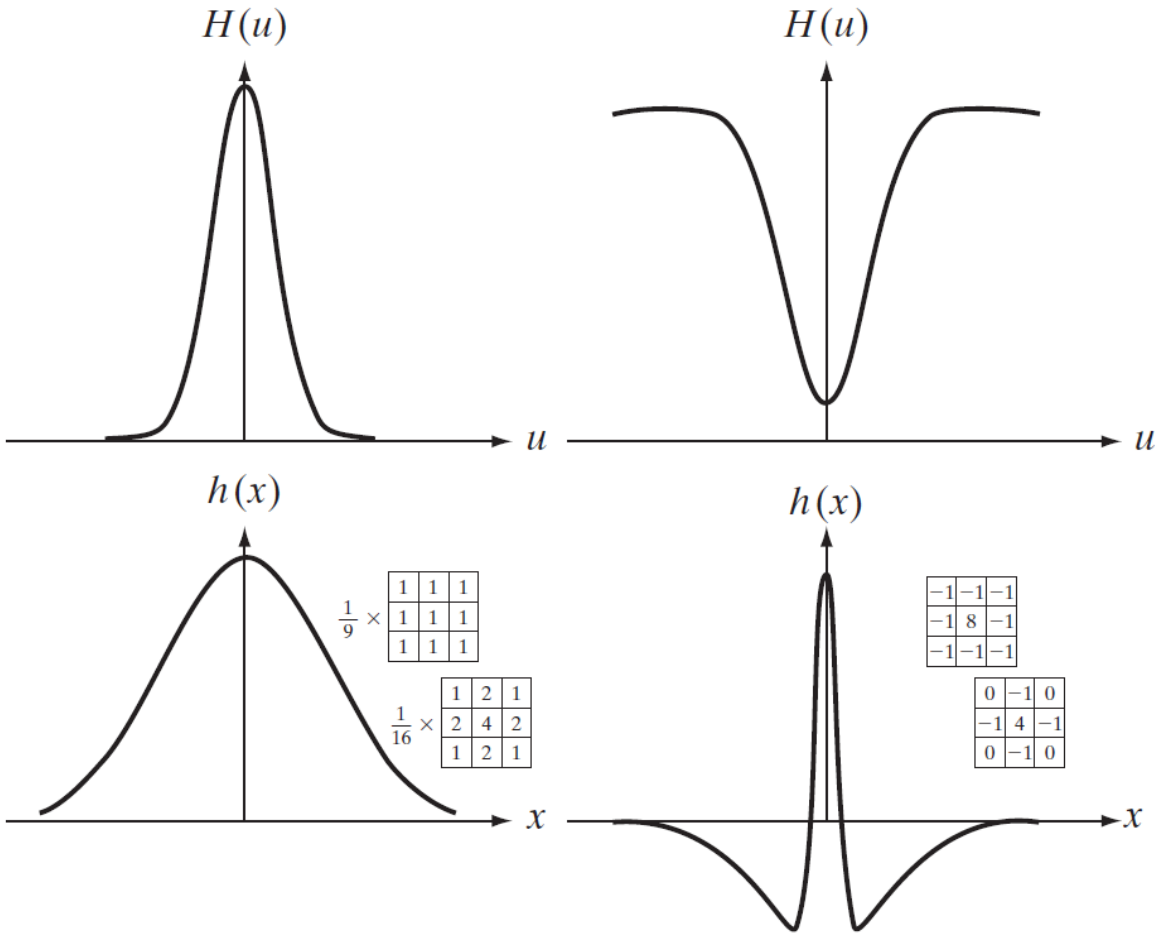
high-pass filtering:



→ sharpening (if added to original)

Gaussian Filters in Frequency Domain

low-pass filter

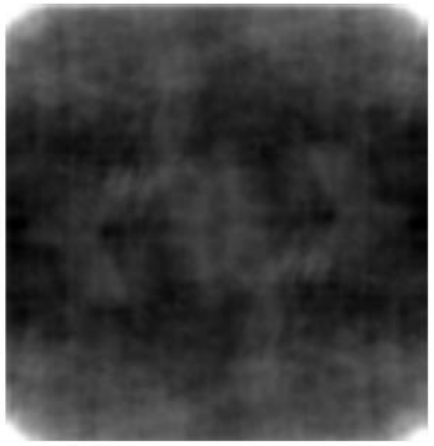


high-pass filter

Importance of Phase



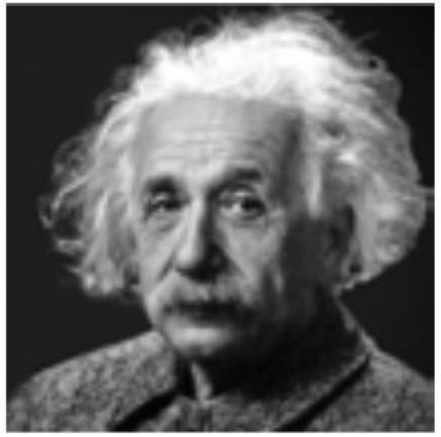
Original Image

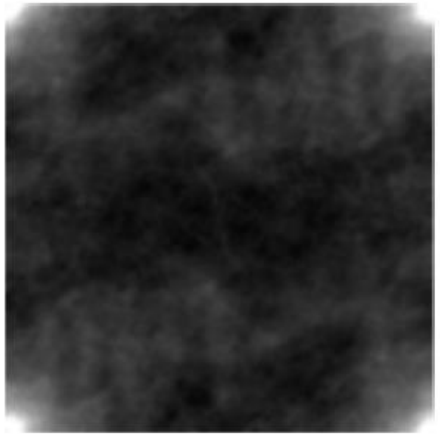


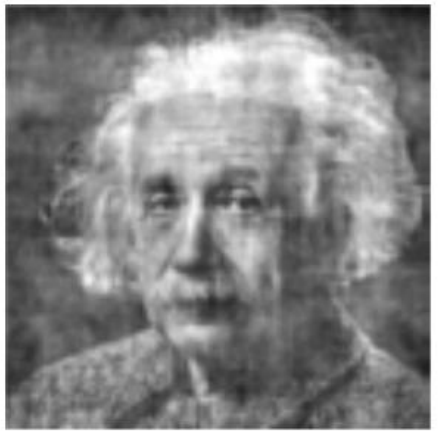
Magnitude Preserved,
Phase Set to Zero



Phase Preserved,
Magnitude Set to Average
of Natural Images







Convolution Theorem

spatial domain:		frequency domain:
$g(x) = f(x) * h(x)$	$\leftrightarrow$	$G(\omega) = F(\omega) \cdot H(\omega)$
$g(x) = f(x) \cdot h(x)$	$\leftrightarrow$	$G(\omega) = F(\omega) * H(\omega)$

Convolution via Fourier Transform

$$g(x) = f(x) * h(x)$$

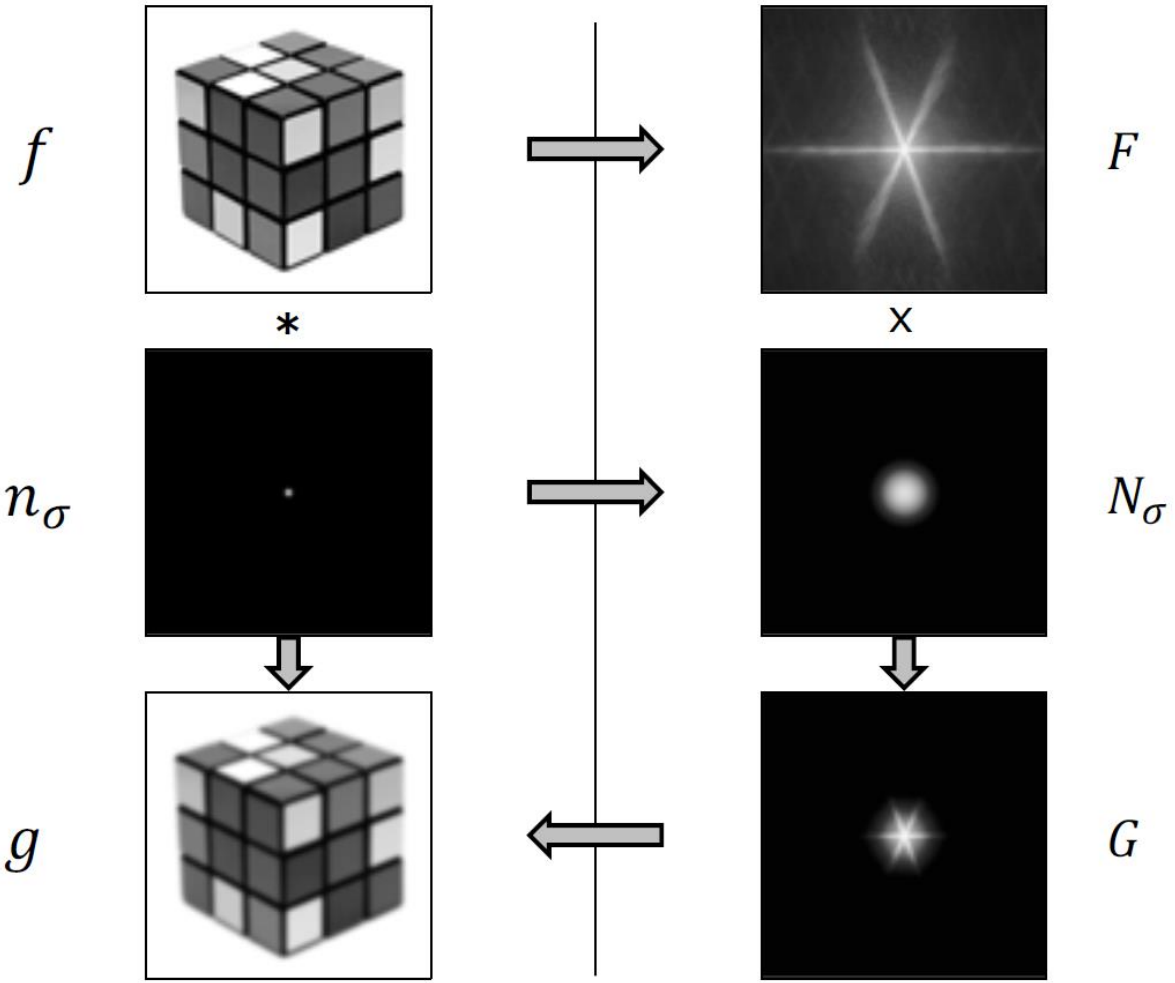
$$\mathcal{F}^{-1}\uparrow \qquad \downarrow\mathcal{F} \qquad \downarrow\mathcal{F}$$

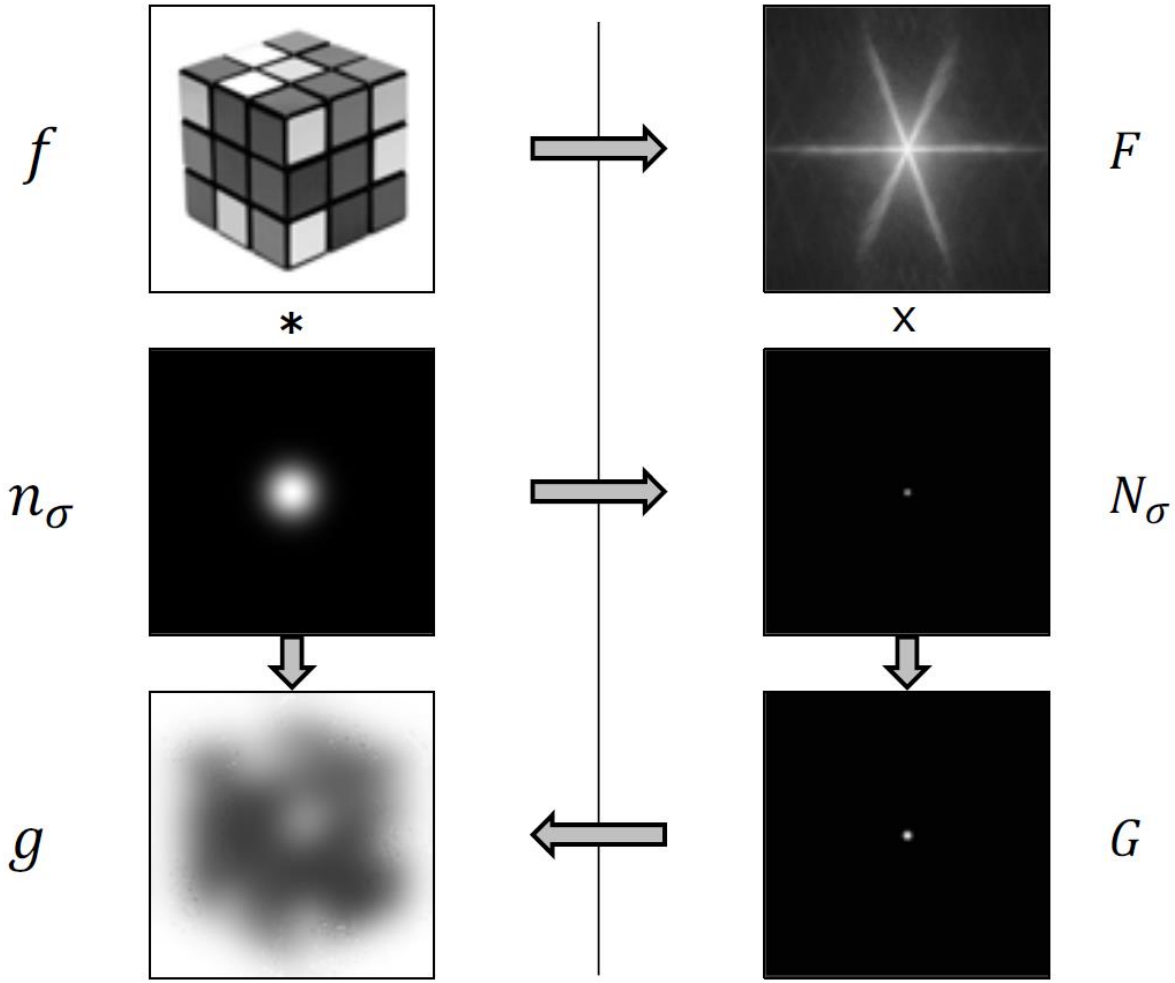
$$G(\omega) = F(\omega) \cdot H(\omega)$$

convolution of images with large spatial filters can be computationally heavy
→ more efficient in frequency domain, thanks to Fast Fourier Transform (enabling convolution times independent of filter size)

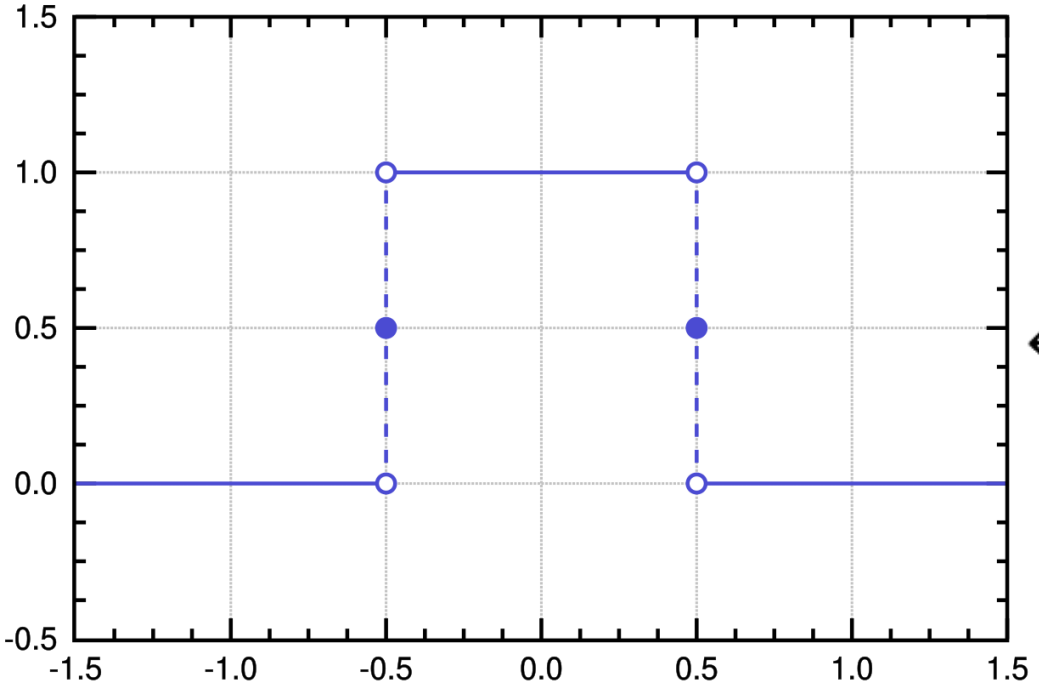
also, construction of filters often easier in frequency domain

Example: Gaussian Smoothing



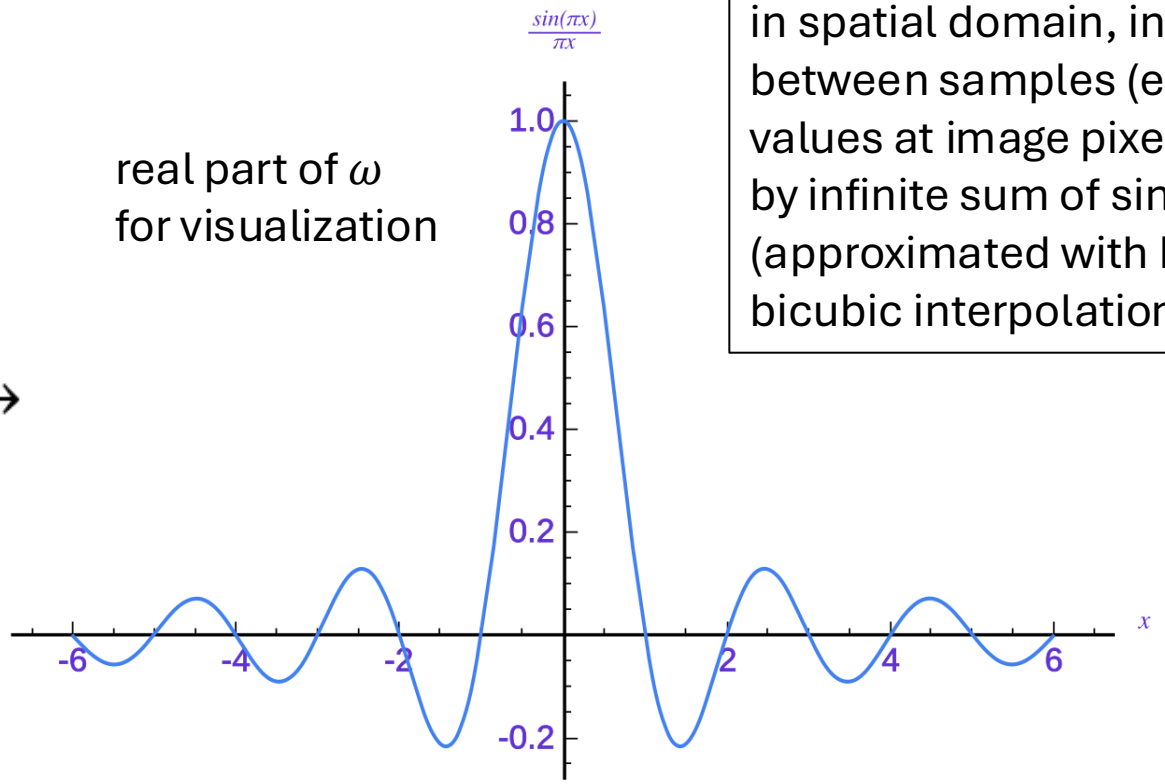


Fourier Transform Pair: Box and sinc Function



↔

real part of ω
for visualization

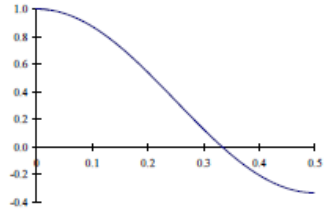
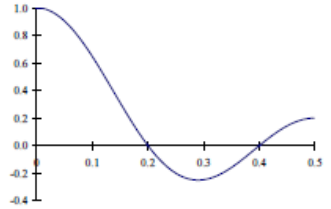
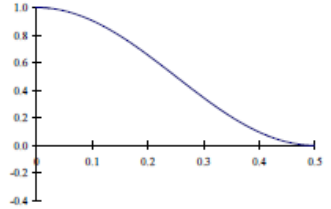
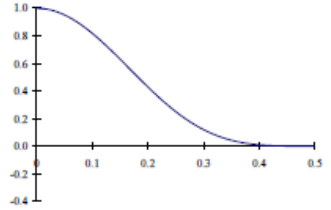
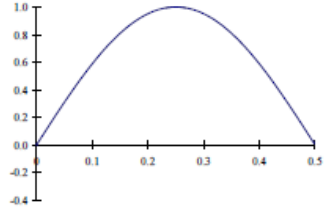
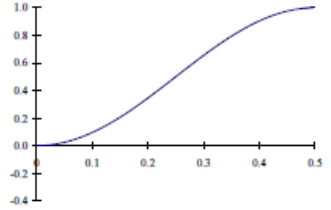


in spatial domain, interpolations between samples (e.g., intensity values at image pixels) formed by infinite sum of sinc functions (approximated with bilinear or bicubic interpolations)

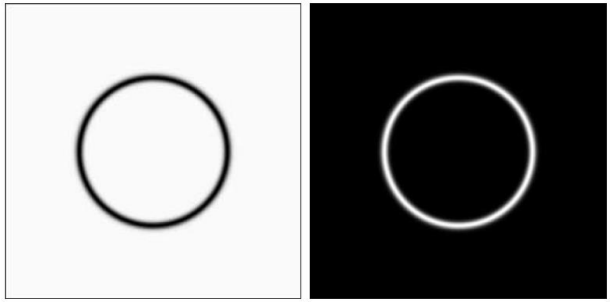
inverse scaling (valid for any function):
wider in spatial domain → more compressed in frequency domain (and vice versa)
(e.g., narrow and broad Gaussian)

smoothing filters / low-pass filters

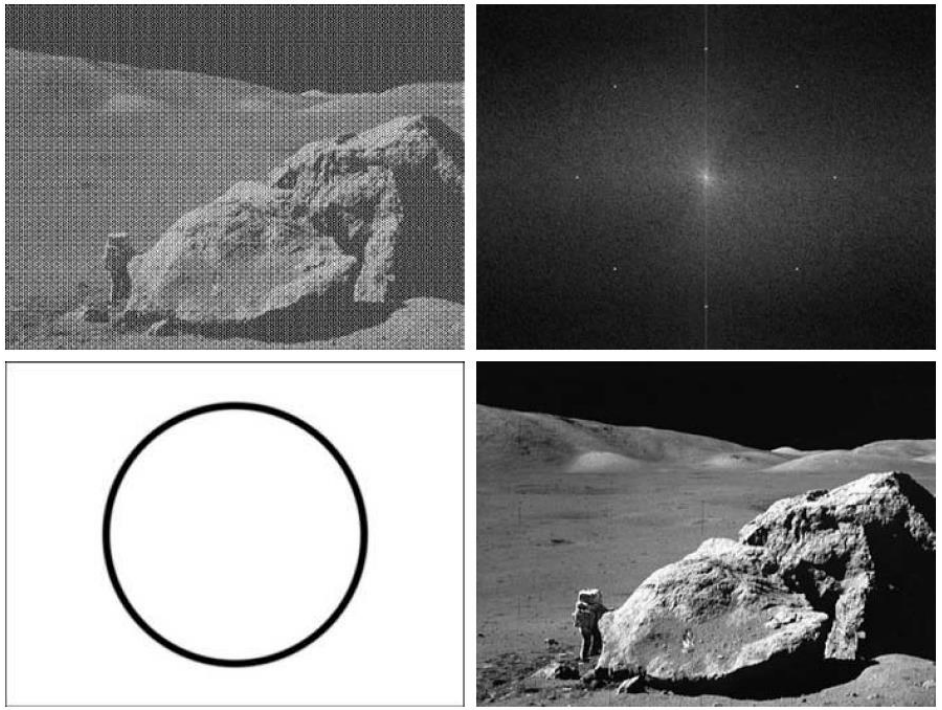
sharpening filters / high-pass filters

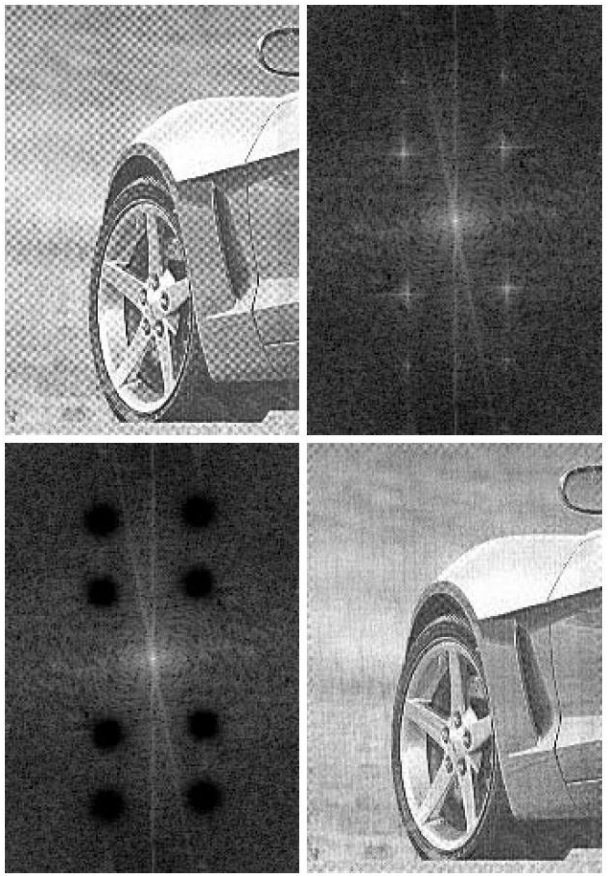
Name	Kernel	Transform	Plot	
box-3	$\frac{1}{3}[1\ 1\ 1]$	$\frac{1}{3}(1+2\cos\omega)$		
box-5	$\frac{1}{5}[1\ 1\ 1\ 1\ 1]$	$\frac{1}{5}(1+2\cos\omega+2\cos 2\omega)$		approximated sinc
linear	$\frac{1}{4}[1\ 2\ 1]$	$\frac{1}{2}(1+\cos\omega)$		
binomial	$\frac{1}{16}[1\ 4\ 6\ 4\ 1]$	$\frac{1}{4}(1+\cos\omega)^2$		approximated Gaussian
Sobel	$\frac{1}{2}[-1\ 0\ 1]$	$\sin\omega$		some smoothing
corner	$\frac{1}{2}[-1\ 2\ -1]$	$\frac{1}{2}(1-\cos\omega)$		

Band-Reject and Band-Pass Filters



periodic noise can be effectively removed with band-reject filters in frequency domain





notch reject filter

Deconvolution

application of convolution theorem: inverse filtering in frequency domain to get rid of degradation effect H

(image restoration)

more complicated in practice:

- degradation function H usually not known completely → need for estimation
- usually also additive noise on top → need for noise suppression during deconvolution (Wiener filter)

example: motion blur

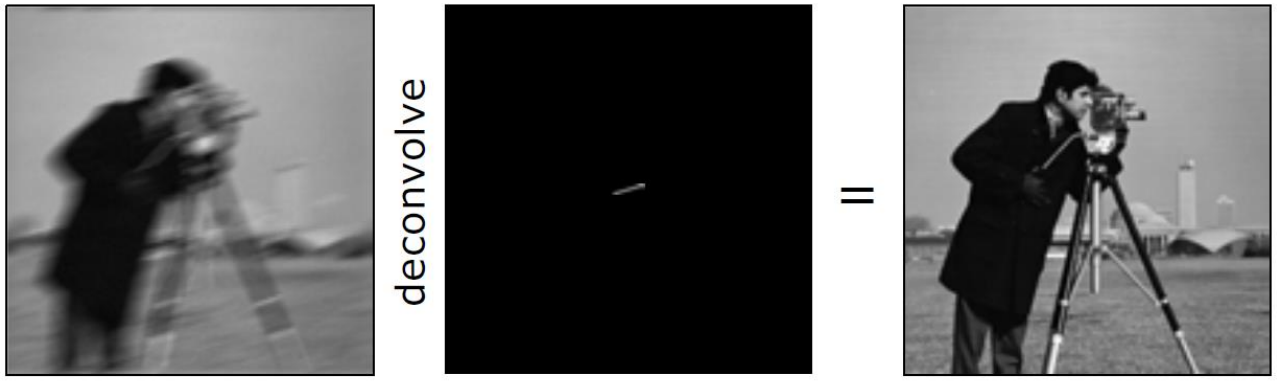


Image $g(x,y)$ PSF $h(x,y)$ Recovered $f'(x,y)$

$$F'(u,v) = \frac{G(u,v)}{H(u,v)} \rightarrow \boxed{\text{IFT}} \rightarrow f'(x,y)$$

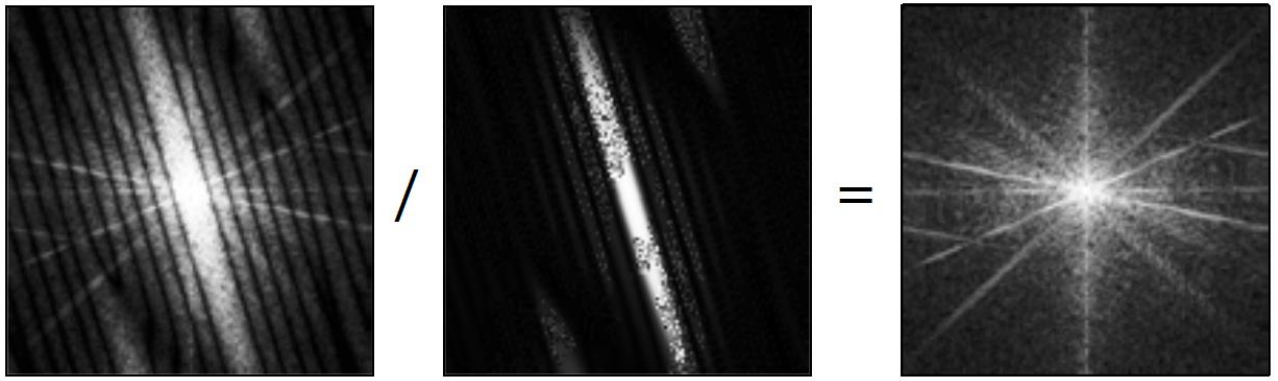
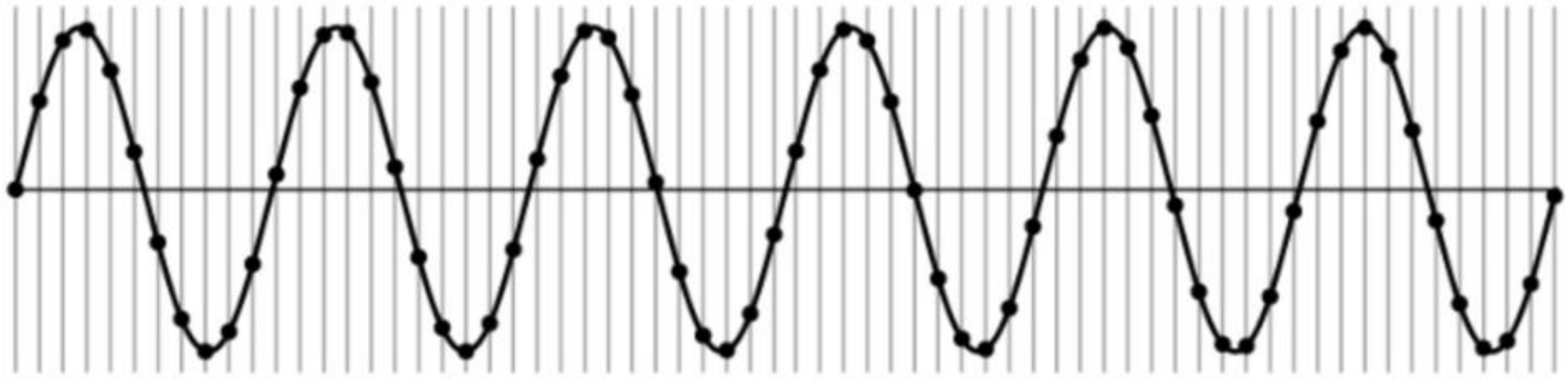


Image $G(u,v)$ PSF $H(u,v)$ Recovered $F'(u,v)$

point spread function

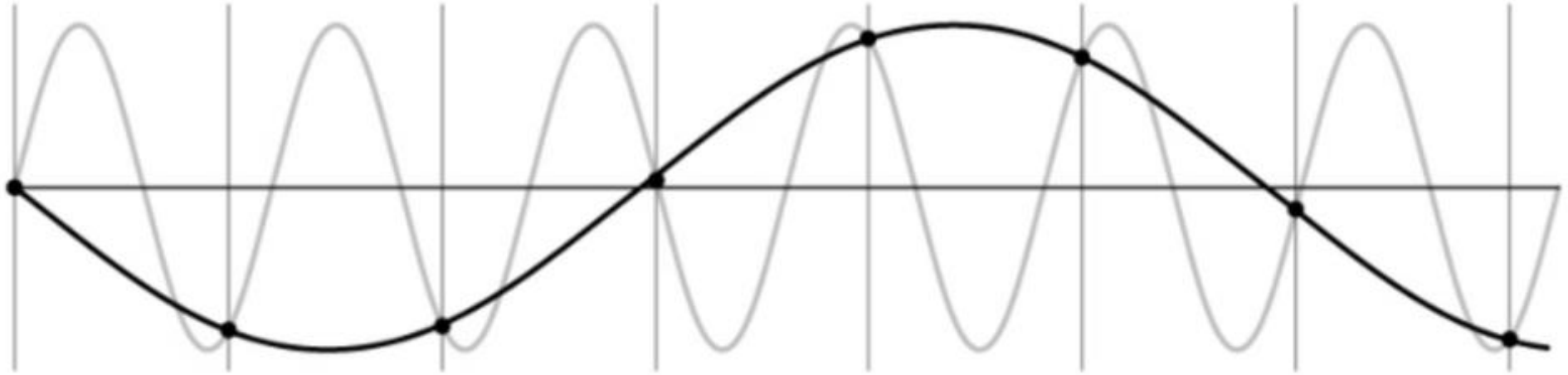
Aliasing

Sampling



remember: digital images sample intensity values at each pixel

Undersampling



aliasing: higher-frequency signal disguised as one of lower frequency

Moiré Effect



aliasing in digital images often manifests itself as ringing patterns

temporal aliasing can be observed as wagon-wheel effect (apparent backward motion) in image sequences (video): frame rate too low with respect to speed of wheel rotation

Sampling Theorem

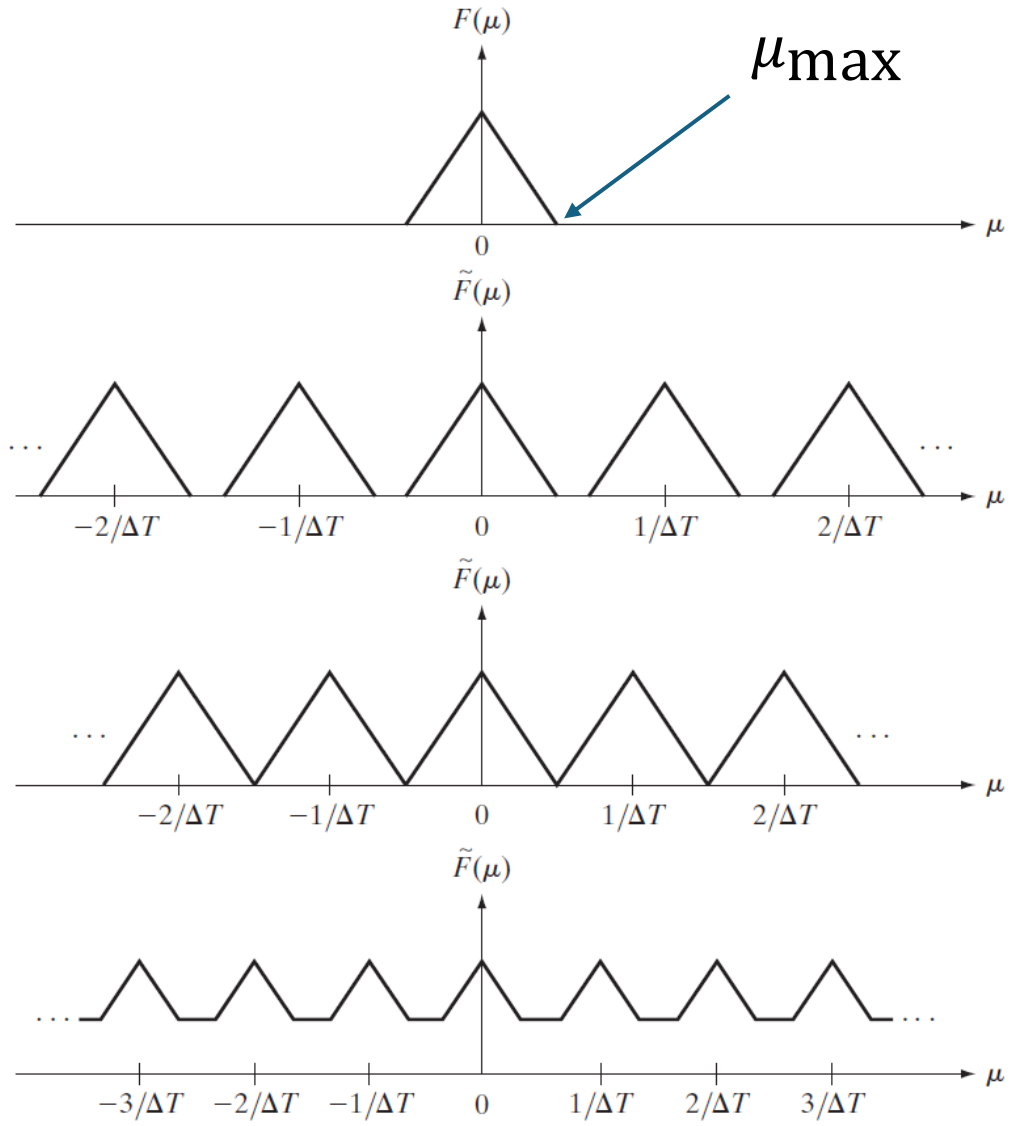
continuous signal can be recovered from its samples, if sampling rate exceeds Nyquist frequency:

$$\frac{1}{\Delta T} > 2\mu_{\max}$$

→ no aliasing

digital images: no aliasing if $\frac{N}{2} > \mu_{\max}$,
with number of pixels N
(second dimension accordingly)

Fourier transform of band-limited function:



Anti-Aliasing in Digital Images

no problem if signal is oversampled: number of pixels higher than twice the maximum frequency of image (for both width and height)

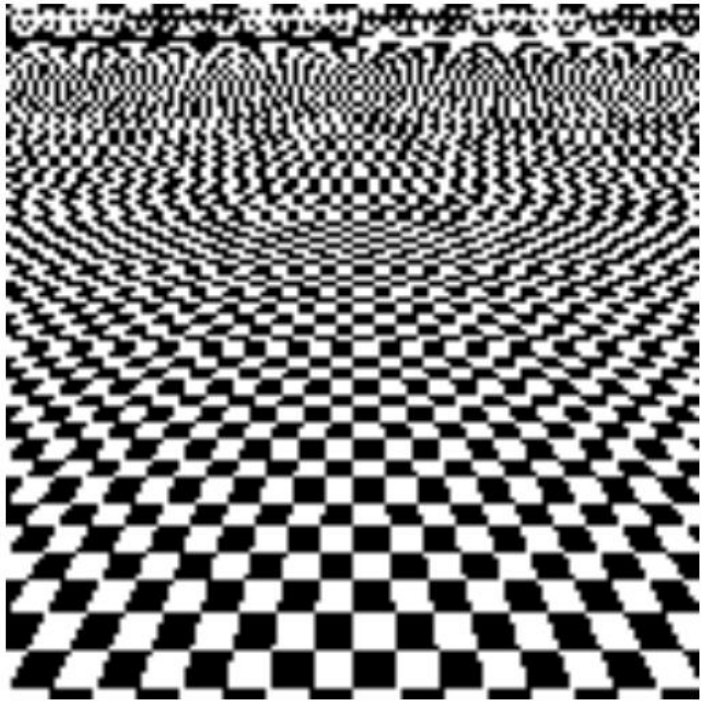
if not oversampled: smooth the signal

→ removing some details (i.e., high frequencies,) that cause aliasing

optical anti-aliasing filter in digital cameras (on top of image sensor): cut out frequencies above Nyquist frequency

Example: Shrinking

smoothing crucial before subsampling (shrinking) to avoid aliasing: decreases Nyquist frequency of image (to counter less pixels)



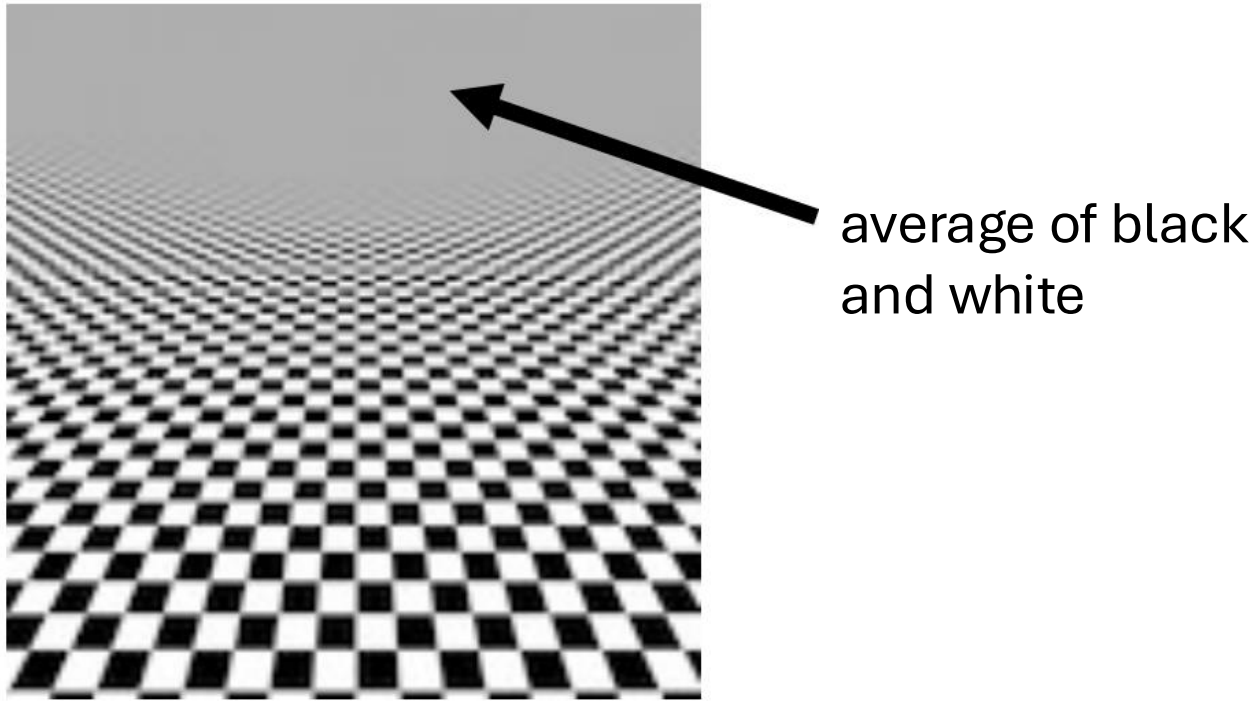
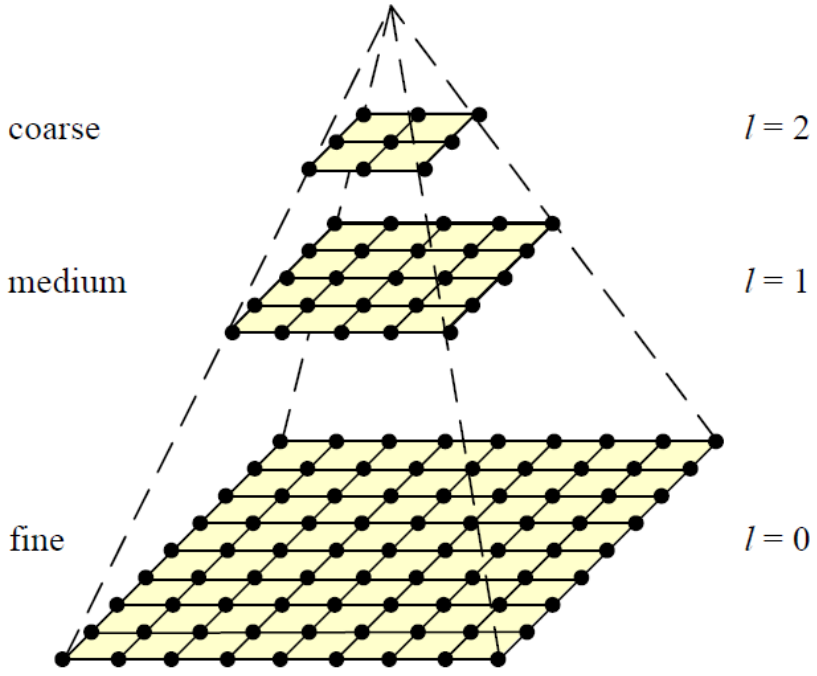


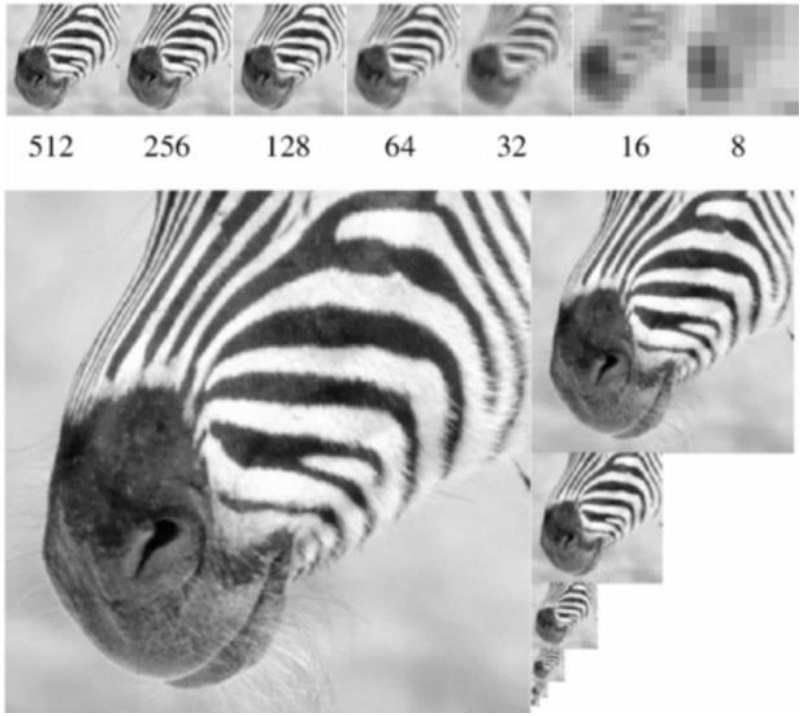
Image Pyramids

Gaussian Image Pyramid

sequence of sub-sampled images (one use case: features at different scales)

apply smoothing before each down-sample step (to avoid aliasing), typically by convolution with binomial kernel (converging to Gaussian)

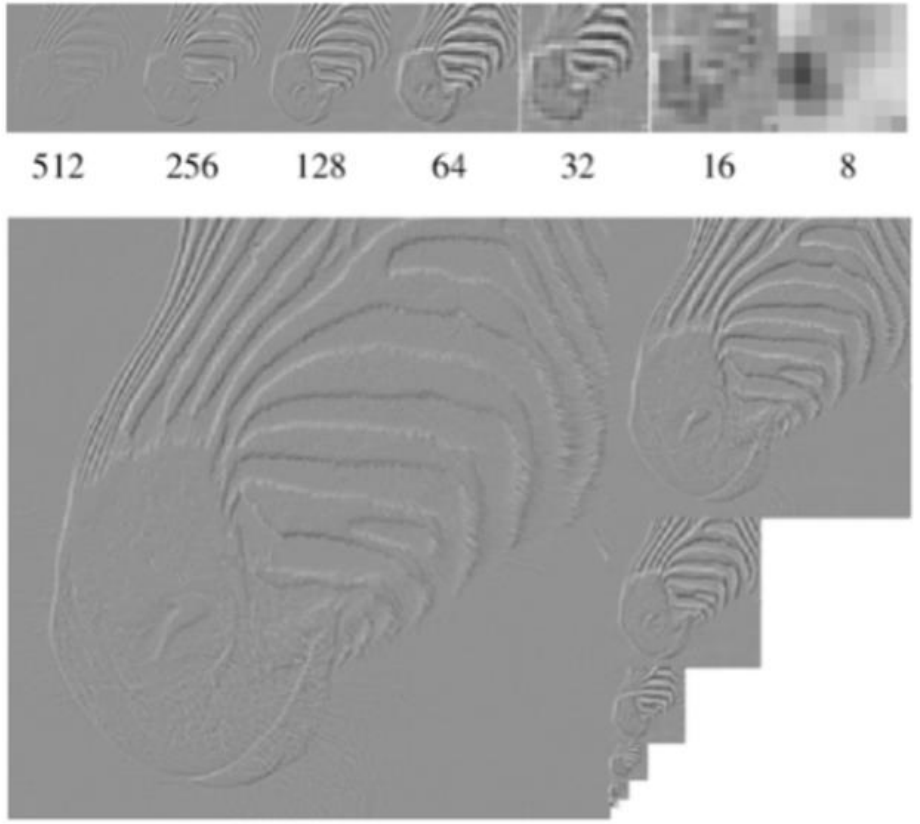




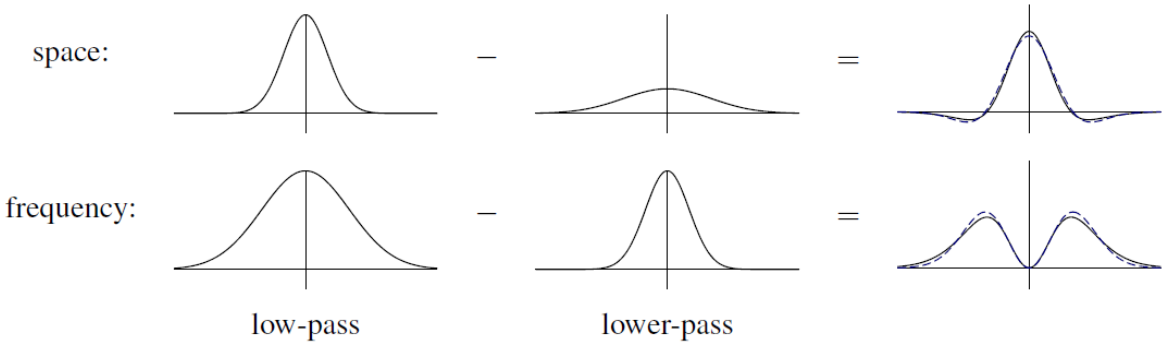
Laplacian Image Pyramid

retain only residuals at different levels (allows later upsampling, starting from highest-level image)





difference of Gaussians approximate Laplacian:



Wavelets

discrete wavelet transform:

- mapping of discrete function (e.g., a digital image) to sequence of coefficients (of a set of basis functions)
- performing sub-band coding (decomposition of an image into set of band-limited components) combined with progressive down-sampling (similar to residual pyramid)



